US Patent & Trademark Office
Patent Public Search | Text View

United States Patent	12397485
Kind Code	B2
Date of Patent	August 26, 2025
Inventor(s)	Tanaka; Takehiko et al.

Camera module and photosensitive assembly thereof

Abstract

A photosensitive assembly includes a circuit board and a photosensitive chip, as well as a molded base. The molded base is integrally formed on the circuit board and the photosensitive chip, and forms a light window for providing a light path for the photosensitive chip. For a portion of the molded base corresponding to a first end side of the molded base adjacent to a flexible region, a distance between outer and inner edges thereof is a; for a portion of the molded base corresponding to an opposite second end side of the molded base away from the flexible region, a distance between outer and inner edges thereof is c, wherein 0.2 mm≤a≤1 mm, and 0.2 mm≤c≤1.5a. The dimensions a and c enable a corresponding forming groove in a molding process to be filled with molding material.

Inventors: **Tanaka; Takehiko (Nara, JP), Huang; Zhen (Zhejiang, CN), Zhao; Bojie (Zhejiang, CN), Mei; Zhewen (Zhejiang, CN)**

Applicant: **Ningbo Sunny Opotech Co., Ltd.** (Zhejiang, CN)

Family ID: **1000008777543**

Assignee: **NINGBO SUNNY OPOTECH CO., LTD (Zhejiang, CN)**

Appl. No.: **17/877391**

Filed: **July 29, 2022**

Prior Publication Data

Document Identifier	Publication Date
US 20220362978 A1	Nov. 17, 2022

Foreign Application Priority Data

CN	201710810764.0	Sep. 11, 2017
CN	201721159970.1	Sep. 11, 2017

Related U.S. Application Data

division parent-doc US 16645537 US 11433584 WO PCT/CN2018/104915 20180911 child-doc US 17877391

Publication Classification

Int. Cl.: **H05K1/18** (20060101); **B29C45/14** (20060101); **B29C45/26** (20060101); **H01L21/56** (20060101); **H01R43/24** (20060101); **H04N23/55** (20230101); **H05K1/02** (20060101); **H05K3/28** (20060101); **H05K3/46** (20060101); **H10F39/12** (20250101); **H10H20/80** (20250101)

U.S. Cl.:

CPC **B29C45/26** (20130101); **B29C45/14655** (20130101); **H01L21/56** (20130101); **H01L21/561** (20130101); **H01R43/24** (20130101); **H04N23/55** (20230101); **H05K1/0274** (20130101); **H05K1/181** (20130101); **H05K3/284** (20130101); **H05K3/4691** (20130101); **H10F39/12** (20250101); **H10H20/80** (20250101); Y10T428/31 (20150115)

Field of Classification Search

CPC: H05K (1/0274); H05K (1/18); H05K (1/181-187); H05K (3/284); H05K (3/4691); H04N (23/54-57); H04N (5/225); H04N (5/2252); H04N (5/2257-2258); H04N (5/232); H01L (21/56); H01L (21/561)

USPC: 361/749-750; 361/782-784; 361/795; 361/803; 174/254; 174/260-263; 348/335; 348/340; 348/357; 348/373-376

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
7166907	12/2006	Onishi	257/434	H04N 23/55
9241097	12/2015	Tam	N/A	H01L 27/14618
2004/0189862	12/2003	Gustavsson et al.	N/A	N/A
2006/0219885	12/2005	Kinoshita	N/A	N/A
2008/0252771	12/2007	Wu	348/E5.025	H04N 23/55
2015/0264290	12/2014	Happoya	29/832	H04N 23/54
2016/0150133	12/2015	Suzuki	348/376	H04N 23/51
2016/0191767	12/2015	Otani	29/832	H04N 23/54
2017/0244877	12/2016	Wang	N/A	G02B 3/0075

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
205961279	12/2006	CN	N/A

103207502	12/2012	CN	N/A
105681637	12/2015	CN	N/A
105681640	12/2015	CN	N/A
105704354	12/2015	CN	N/A
105847645	12/2015	CN	N/A
206302476	12/2016	CN	N/A
207744032	12/2017	CN	N/A

OTHER PUBLICATIONS

International Search Report issued Nov. 29, 2018 in International (PCT) Application No. PCT/CN2018/104915. cited by applicant

---

Primary Examiner: Dinh; Tuan T

Attorney, Agent or Firm: Wenderoth, Lind & Ponack, L.L.P.

Background/Summary

FIELD OF THE INVENTION

(1) The present application relates to field of camera modules, and further relates to a photosensitive assembly manufactured by a molding process and a manufacturing method thereof, and a camera module and an electronic apparatus having the photosensitive assembly, and a molding die thereof.

BACKGROUND OF THE INVENTION

(2) The molding and packaging technology of a camera module is a newly developed packaging technology based on traditional COB packaging. As shown in FIGS. **1**A to **1**C, a circuit board is packaged by using an existing integrated packaging technology. In this structure, a packaging portion **1** is packaged to a circuit board **2** and a photosensitive chip **3** by an integrated packaging way to form an integrated packaging assembly, and the packaging portion **1** covers a plurality of electronic components **201** of the circuit board **2** and a series of leads **202** electrically connecting the photosensitive chip **3** and the circuit board **2**, so as to enable the length, width, and thickness dimensions of the camera module to be reduced, and assembly tolerances to be reduced. A lens or a lens assembly above the integrated packaging assembly can be installed flat, and the problem that the dust attached to the electronic components affects imaging quality of the camera module can be solved.

(3) More specifically, as shown in FIGS. **1**A and **1**B, in order to improve production efficiency, the integrated packaging assembly is generally produced by a jointed panel production way, that is, a plurality of the integrated packaging assemblies are produced at one time. More specifically, FIGS. **1**A and **1**B show a way of producing the integrated packaging assembly by using a molding die to perform jointed panel. Wherein, the molding die includes an upper die **101** and a lower die **102**, wherein one of the circuit board jointed panels is placed in the lower die **102** of the molding die, the circuit board jointed panel includes a plurality of rows of circuit boards, each row of the circuit boards includes a plurality of circuit boards **2**, and each of the circuit board **2** is operatively connected to the photosensitive chip **3**. The upper die **101** and the lower die **102** are clamped to form a molding cavity, so that the upper die **101** is pressed against the circuit board jointed panel, corresponding to two end sides of the photosensitive chip **3** on each row of the circuit boards, two flow passages **103** and **104** are formed in the upper die, and the upper die **101** has a plurality of bumps **105**, an intermediate flow passage **106** is formed between two adjacent bumps **105**, so that

the plurality of intermediate flow passages **106** extend between the two flow passages **103** and **104**. In the molding process, fluid-like packaging material **4** flows forward along with the two flow passages **103** and **104**, and fills into the intermediate flow passage **106** between two adjacent bumps **105**, so that the region between two adjacent photosensitive chips **3** is also filled with the packaging material **4**, thereby after the packaging material **4** is cured, the packaging portion **1** can be formed on the corresponding respective circuit boards **2** and respective photosensitive chips **3**, a light window located in the middle of the packaging portion **1** is formed at a position corresponding to respective bumps **105**, and these packaging portions **1** are integrally molded to form a one-piece structure, as shown in FIG. **1**C.

(4) Referring to FIG. **1**F, thermosetting packaging material **4** has a curing time T in the molding process, as time passes, its viscosity decreases to the lowest point, and then gradually rises to the highest point to be completely cured. Ideally, when the packaging material **4** has a low viscosity, the packaging material **4** is to fill the flow passages **103**, **104**, and **106**, while the packaging material **4** still flows forward when the viscosity is large, the friction of the lead **202** between the circuit board **2** and the photosensitive chip **3** is relatively large, which may easily cause deformation and damage of the lead **202**.

(5) In the above-mentioned molding process, the packaging material **4** is a thermosetting material, and after melting, enters the two flow passages **103** and **104**, and is cured under the action of heating conditions. However, in actual production, it has been found that during the molding process, when the packaging material **4** flows forward along with the two flow passages **103** and **104**, if the widths of the two flow passages **103** and **104** are not the same, the ratio of the width thereof is not within a reasonable range, it will cause the packaging material **4** to move inconsistently in the forward flow process.

(6) More specifically, because the packaging material **4** is a fluid having a predetermined viscosity, the dimensions of the two flow passages **103** and **104** are relatively small and assuming that the flow passage **103** is a narrow flow passage, the flow rate in the flow passage **103** is relatively small, and the friction generated by the inner wall of the flow passage **103** on the fluid-like packaging material **4** in the flow passage **103** has a relatively large effect on flow velocity of the fluid-like packaging material **4**, so the flow velocity of the packaging material **4** in the flow passage **103** is relatively slow. The flow passage **104** is a relatively wide flow passage, the flow rate therein is relatively large, and the friction generated by the inner wall of the flow passage **104** on the fluid-like packaging material **4** in the flow passage **104** has a relatively small effect on flow velocity of the fluid-like packaging material **4**, so that the flow velocity of the packaging material **4** in the flow passage **104** is relatively fast. In this way, within the curing time T of the packaging material **4**, the packaging material **4** in the flow passage **104** can flow from the feeding end to the terminal end, and at least portion of the intermediate flow passage **106** is filled with the packaging material **4**. However, the packaging material **4** in the flow passage **103** may not be able to flow from the feeding end to the terminal end within the curing time T, so that the local position of the flow passage **103** cannot be filled, as the region S shown in FIG. **1**D, thereby it is impossible to form a one-piece structure of the packaging portion **1** with a series of complete shapes between the upper die **101** and the lower die **102**, at the position corresponding to the region S, the packaging portion **1** forms a gap, so that it is not possible to form an all-around closed light window.

(7) In addition, if the flow velocity of the packaging material **4** in the above two flow passages **103** and **104** is not consistent, the flow velocity in the flow passage **104** is faster during this curing time T, and the flow passage **104** and the flow passage **106** can be separately filled with the packaging material **4** when the viscosity is low, and the packaging material **4** in the flow passage **103** flows too slowly forward, when the viscosity is relatively large, it still flows forward in the flow passage **103**, resulting in the frictional force on the lead **202** through which the packaging material flow is large, so that the lead **202** is deflected forward with a large radius, thereby the lead **202** is easily deformed and damaged, and is easily detached from a bonding pad.

(8) Another situation is that when the packaging material **4** reaches the region between the first photosensitive chip and the second photosensitive chip, more packaging material **4** is filled into the region through the flow passage **104**, so that as the packaging material **4** flows forward along with the two flow passages **103** and **104**, which will cause the ratio of the packaging material **4** of the flow passage **104** to be filled between the two adjacent photosensitive chips **3** to gradually increase. Eventually, the packaging material **4** of the flow passage **104** may flow to the flow passage **103** and prevented the packaging material **4** in the flow passage **103** from further flow forward. Thus, eventually, the packaging material **4** in the flow passage **103** may not flow to the terminal end of the flow passage **103** after the packaging material **4** is cured, so that the entire molding cavity cannot be filled with the packaging material **4**, thereby a complete ring-shaped packaging portion **1** is not provided for each of the corresponding circuit boards **2** and the photosensitive chips **3**. Moreover, the photosensitive chip **3** and the circuit board **2** are connected through the flexible lead **202**, when the packaging material **4** flowing forward in the two flow passages **103** and **104** extends laterally to the intermediate flow passage **106** between two adjacent photosensitive chips, the packaging material **4** flowing from the two flow passages **103** and **104** is basically equivalent to converging in the intermediate flow passage **106** of the same dimension, so that the flow velocity is substantially the same, no turbulence is generated, and deformation and damage of the leads **202** for connecting the photosensitive chip **3** and the circuit board **2** between two adjacent photosensitive chips **3** are not caused. However, when the packaging material **4** in the wider flow passage **104** reaches the flow passage **103** from the intermediate flow passage **106**, the packaging material **4** in the flow passage **103** will be prevented from continuing to flow forward, and because the different dimensions of the flow passage **104** and the flow passage **103**, the turbulence is formed by collision generated between the two fluids at the confluence, such turbulence can easily cause deformation of the lead **202**, or even cause the lead **202** to break, thereby forming a defective product.
(9) More specifically, when the fluid of the flow passage **104** flows into the flow passage **103** and flows reversely toward the feeding end, as shown in FIG. **1**E, when the packaging material **4** in the two flow passages **103** and **104** flows face to face in opposite directions and produce turbulence, there is a pressure difference between the two fluids, causing the leads **202** through which the two fluids flow to swing in different directions, so friction between adjacent leads **202** may cause damage to the lead **202**.
(10) In addition, when the width dimensions of the two flow passages **103** and **104** are sufficiently large, within the curing time T, the packaging material **4** can fill the flow passages **103**, **104**, and **106**, but this will result in a larger dimension of the packaging portion **1** finally formed corresponding to each circuit board **2**, so that it does not meet the requirements of some miniaturized camera modules.

SUMMARY OF THE INVENTION

(11) An object of the present application is to provide a camera module, a photosensitive assembly thereof and a manufacturing method, wherein in a method for manufacturing a jointed panel of a photosensitive assembly, a molding material can fill a base jointed panel molding guide groove in a molding die in a molding process to avoid occurrence of defective products of the photosensitive assembly.
(12) An object of the present application is to provide a camera module, a photosensitive assembly thereof and a manufacturing method, wherein in a molding process, the molding material can form a one-piece molding base on a circuit board assembly, and the one-piece molding base can form an all-round closed light window at positions corresponding to each photosensitive assembly, so that after the formed one-piece jointed panel of the photosensitive assembly is cut, a molding base having the light window is formed on each circuit board and the corresponding photosensitive assembly to prevent a portion of the molding base from forming an opening to connect the light window to outside of the molding base.

(13) An object of the present application is to provide a camera module, a photosensitive assembly thereof, and a manufacturing method, wherein the base jointed panel molding guide groove is used to form the one-piece molding base on a row of circuit boards, which has two diversion grooves of two sides, and a plurality of filling grooves extending laterally between the two diversion grooves, the molding material flows and is cured in the diversion grooves and the filling grooves, wherein a dimension ratio between two diversion grooves is within a preset range, so that the molding material can flow forward from feeding ends of the two diversion grooves and fill the entire diversion groove and filling groove of the base jointed panel molding guide grooves.
(14) An object of the present application is to provide a camera module, a photosensitive assembly thereof, and a manufacturing method, wherein the base jointed panel molding guide groove is used to form the one-piece molding base on two rows of adjacent circuit boards integrated in a rigid region, and has two first diversion grooves on both sides, and a second diversion groove in the middle, and a plurality of filling grooves located between the two first diversion grooves and the second diversion groove, the molding material flows and is cured in the diversion groove and the filling groove, wherein a dimension ratio between the two first diversion grooves and the second diversion groove is in a preset range, so that the molding material can flow forward from feeding ends of the two diversion grooves and fill the entire diversion groove and filling groove of the base jointed panel molding guide groove.
(15) An object of the present application is to provide a camera module, a photosensitive assembly thereof, and a manufacturing method, wherein when the dimension of the diversion groove is small to form a miniaturized photosensitive assembly, by selecting a dimension ratio relationship between these diversion grooves, the diversion groove with a small dimension is made to still fill the entire base jointed panel molding guide groove during the molding process.
(16) An object of the present application is to provide a camera module, a photosensitive assembly thereof, and a manufacturing method, wherein before viscosity of the molding material reaches a high value and the molding material is cured, the molding material can fill the base jointed panel molding guide groove, thereby preventing connecting wire between the circuit board and the photosensitive assembly from being damaged by a high-viscosity molding material flowing forward.
(17) An object of the present application is to provide a camera module, a photosensitive assembly thereof, and a manufacturing method, wherein the dimension and ratio of the diversion groove are within a preset range, so that the molding material can reaches from the feeding end to the terminal end of each of the above-mentioned diversion grooves in a molding process to prevent the molding material in one diversion groove from flowing to the other diversion groove and obstructing the molding material in the other diversion groove from flowing forward.
(18) An object of the present application is to provide a camera module, a photosensitive assembly thereof, and a manufacturing method, wherein the molding material flowing forward through the diversion grooves converges substantially at the filling grooves in the middle to prevent the molding material in a certain diversion groove from filling the filling groove and further flowing to the other diversion groove to generate a violent collision turbulence, which results in deformation and damage of the connecting wire connecting the circuit board and the photosensitive assembly.
(19) An object of the present application is to provide a camera module, a photosensitive assembly thereof, and a manufacturing method, wherein the molding material flows forward in the diversion groove, thereby preventing the molding material in a certain guide groove form flowing through the filling groove and further flowing to another conducting fluid and flowing backward toward the feeding end, thereby affecting the filling efficiency of the fluid-like molding material, and causing adjacent connecting wires to deflect close to each other, which results in occurrence of friction and damage of the connecting wire.
(20) An object of the present application is to provide a camera module, a photosensitive assembly thereof and a manufacturing method, wherein the molding material can be selected from a material

with a relatively high viscosity range, thereby avoiding at the time of selecting a material with a relatively small viscosity range, the molding material easily entering into the photosensitive region of the photosensitive assembly during the molding process to form a flash, and at the same time, to avoid, for preventing the flash, increasing the pressure exerted by the light window molding portion on the photosensitive assembly to crush the photosensitive assembly.
(21) An object of the present application is to provide a camera module, a photosensitive assembly thereof and a manufacturing method, wherein the molding process can form the one-piece molding base on a row of circuit boards having a plurality of circuit boards and a row of photosensitive assemblies at one time, so that a row of multiple photosensitive assemblies are formed by a jointed panel process, such as preferably 2-12 photosensitive assemblies.
(22) In order to achieve at least one of the above objects of the present application, the present application provides a method for manufacturing a photosensitive assembly of a camera module, comprising the following steps:
(23) (a) fixing a circuit board jointed panel to a second die of a molding die, wherein the circuit board jointed panel includes one or more rows of circuit boards, and each row of circuit boards includes one or more circuit boards arranged side by side, each of the circuit boards includes a combined rigid region and flexible region, and each of the circuit boards is operatively connected with a photosensitive chip;
(24) (b) clamping the second die and a first die and filling molten molding material in a base jointed panel molding guide groove in the molding die, wherein a position corresponding to a light window molding portion is prevented from being filled with the molding material; and
(25) (c) curing the molding material in the base jointed panel molding guide groove to form a one-piece molding base at a position corresponding to the base jointed panel molding guide groove, wherein the one-piece molding base is integrally molded on corresponding each row of the circuit boards and each row of the photosensitive chips to form a photosensitive assembly jointed panel and form a light window for providing a light path for each of the photosensitive chips at a position corresponding to the light window molding portion, wherein the base jointed panel molding guide groove has a first diversion groove corresponding to a first end side of the one-piece molding base adjacent to the flexible region and a second diversion groove corresponding to the one-piece molding base away from the flexible region, and a plurality of filling grooves extending between the first diversion groove and the second diversion groove, wherein each of the light window molding portions is located between two adjacent filling grooves, wherein a width of a bottom end of the first diversion groove is a, and a width of a bottom end of the second diversion groove is c, wherein the width a corresponds to a distance between an outer edge and an inner edge of a portion of the one-piece molding base of the first end side of the one-piece molding base adjacent to the flexible region; wherein the width c corresponds to a distance between an outer edge and an inner edge of a portion of the one-piece molding base of an opposite second end side of the one-piece molding base away from the flexible region, wherein 0.2 mm≤a≤1 mm, 0.2 mm≤c≤1.5a.
(26) In the above method for manufacturing the photosensitive assembly of the camera module, further including a step of cutting the photosensitive assembly jointed panel to obtain a plurality of photosensitive assemblies, wherein each of the photosensitive assemblies includes the circuit board, the photosensitive chip, and the molding base, wherein the molding base is integrally molded on the circuit board and the photosensitive chip and forms the light window for providing the light path for the photosensitive chip.
(27) In the above method for manufacturing the photosensitive assembly of the camera module, there is 0.7a≤c≤1.3a.
(28) In the above method for manufacturing the photosensitive assembly of the camera module, further including a step of cutting a portion of the photosensitive assembly corresponding to the opposite second end side of the molding base away from the flexible region, so that a distance between an outer edge and an inner edge of the remaining part of the molding base is b, wherein

0.2 mm≤b≤1.5a-0.2 mm.

(29) In the above method for manufacturing the photosensitive assembly of the camera module, the rigid region of each of the circuit boards has a pressing distance W for a pressing block of the first die in the molding process to facilitate pressing on the first end side adjacent to the flexible region, and the pressing distance W has a value ranging from 0.1˜1 mm.

(30) In the above method for manufacturing the photosensitive assembly of the camera module, the rigid regions of each row of the circuit boards are integrally molded to form an integral rigid region jointed panel.

(31) According to another aspect of the present application, the present application provides a method for manufacturing a photosensitive assembly of a camera module, comprising the following steps: (a) fixing a circuit board jointed panel to a second die of a molding die, wherein the circuit board jointed panel includes a plurality of rows of circuit boards, and each row of circuit boards includes one or more circuit boards arranged side by side, each of the circuit boards includes a combined rigid region and flexible region, and each of the circuit boards is operatively connected with a photosensitive chip; (b) clamping the second die and a first die and filling molten molding material in a base jointed panel molding guide groove in the molding die, wherein a position corresponding to a light window molding portion is prevented from being filled with the molding material; and (c) curing the molding material in the base jointed panel molding guide groove to form a one-piece molding base at a position corresponding to the base jointed panel molding guide groove, wherein the one-piece molding base is integrally molded on two adjacent rows of the circuit boards and two adjacent rows of the photosensitive assemblies to form a photosensitive assembly jointed panel and form a light window for providing a light path for each of the photosensitive chips at a position corresponding to the light window molding portion, wherein the two adjacent rows of the circuit boards are arranged such that their flexible regions are away from each other and their rigid regions are adjacent to each other, wherein the base jointed panel molding guide groove has two first diversion grooves corresponding to two end sides of the one-piece molding base adjacent to the flexible region and a second guide groove corresponding to a region between the two adjacent rows of photosensitive chips, and a plurality of filling grooves extending between the first diversion groove and the second diversion groove, wherein each of the light window molding portions is located between two adjacent filling grooves, wherein a width of a bottom end of the first diversion groove is a, and a width of a bottom end of the second diversion groove is c, wherein the width a corresponds to a distance between an outer edge and an inner edge of a portion of the one-piece molding base of the each end side of the one-piece molding base adjacent to the flexible region; wherein the width c corresponds to a distance between two inner edges of a portion of the one-piece molding base extending between the two adjacent rows of the photosensitive chips, wherein 0.2 mm≤a≤1 mm, 0.2 mm≤c≤1.5a.

(32) In the above method for manufacturing the photosensitive assembly of the camera module, further including a step of cutting the photosensitive assembly jointed panel to obtain a plurality of photosensitive assemblies, wherein each of the photosensitive assemblies includes the circuit board, the photosensitive chip, and the molding base, wherein the molding base is integrally molded on the circuit board and the photosensitive chip and forms the light window for providing a light path for the photosensitive chip.

(33) In the above method for manufacturing the photosensitive assembly of the camera module, there is 0.7a≤c≤1.3a.

(34) In the above method for manufacturing the photosensitive assembly of the camera module, further including a step of cutting a portion of the photosensitive assembly located between the two adjacent rows of the photosensitive assemblies to obtain a portion of the molding base corresponding to the opposite another end side of the molding base away from the flexible region, and having a distance between the outer edge and the inner edge thereof as b, where 0.2 mm≤b≤1.5a-0.2 mm.

(35) In the above method for manufacturing the photosensitive assembly of the camera module, the rigid region of each of the circuit boards has a pressing distance W for a pressing block of the first die in the molding process to facilitate pressing on the first end side adjacent to the flexible region, and the pressing distance W has a value ranging from 0.1˜1 mm.
(36) In the above method for manufacturing the photosensitive assembly of the camera module, the rigid regions of the two adjacent rows of the circuit boards are integrally molded to form an integral rigid region jointed panel.
(37) According to still another aspect of the present application, the present application provides a photosensitive assembly of a camera module, comprising: a circuit board, including a combined rigid region and flexible region; a photosensitive chip; and a molding base, wherein the molding base is integrally molded on the circuit board and the photosensitive chip and forms a light window for providing a light path for the photosensitive chip; wherein a distance between an outer edge and an inner edge of a portion of the molding base corresponding to a first end side of the molding base adjacent to the flexible region is a; a distance between an outer edge and an inner edge of the portion of the molding base corresponding to an opposite second end side of the molding base away from the flexible region is c, wherein 0.2 mm≤a≤1 mm, 0.2 mm≤c≤1.5a.
(38) In the above photosensitive assembly of the camera module, there is 0.7a≤c≤1.3a.
(39) In the above photosensitive assembly of the camera module, a portion of the photosensitive assembly away from the opposite second end side of the flexible region is adapted to be cut, so that a distance between an outer edge and an inner edge of the remaining part of the molding base after being cut is b, wherein 0.2 mm≤b≤1.5a-0.2 mm.
(40) In the above photosensitive assembly of the camera module, the circuit board further includes a plurality of electronic components, and wherein the plurality of electronic components are provided at at least one wing side on both sides of the photosensitive assembly on the rigid region except for both end sides being adjacent to the flexible region and away from the flexible region, and the molding base integrally embeds the electronic component.
(41) In the above photosensitive assembly of the camera module, at the first end side of the molding base adjacent to the flexible region, a pressing distance W for a pressing block of the molding die in the molding process to facilitate pressing on the rigid region is between the outer edge of the molding base and the outer edge of the rigid region, and the pressing distance W has a value ranging from 0.1˜1 mm.
(42) In the above photosensitive assembly of the camera module, further including a filter assembly and a filter assembly lens holder, wherein the filter assembly is assembled to the filter assembly lens holder, and the filter assembly lens holder is assembled on a top side of the molding base.
(43) In the above photosensitive assembly of the camera module, further including a protective frame provided on the photosensitive chip, and the molding base is integrally molded on the circuit board, the photosensitive chip and the protective frame.
(44) According to still another aspect of the present application, the present application provides a photosensitive assembly of a camera module, comprising: a circuit board, including a combined rigid region and flexible region; a photosensitive chip; and a molding base, wherein the molding base is integrally molded on the circuit board and the photosensitive chip and forms a light window for providing a light path for the photosensitive chip; wherein a distance between an outer edge and an inner edge of a portion of the molding base corresponding to a first end side of the molding base adjacent to the flexible region is a; a portion of the molding base corresponding to an opposite second end side of the molding base away from the flexible region has a cutting surface, and a distance between an outer edge and an inner edge thereof is b, wherein 0.2 mm≤a≤1 mm, 0.2 mm≤b≤1.5a-0.2 mm.
(45) In the above photosensitive assembly of the camera module, the circuit board further includes a plurality of electronic components, and the plurality of electronic components are provided at at least one wing side on both sides of the photosensitive assembly on the rigid region except for both

end sides being adjacent to the flexible region and away from the flexible region, wherein the molding base integrally embeds the electronic component.

(46) In the above photosensitive assembly of the camera module, at the first end side of the molding base adjacent to the flexible region, a pressing distance W for a pressing block of the molding die in the molding process to facilitate pressing on the rigid region is between an outer edge of the molding base and an outer edge of the rigid region, and the pressing distance W has a value ranging from 0.1˜1 mm.

(47) In the above photosensitive assembly of the camera module, further including a filter assembly and a filter assembly lens holder, wherein the filter assembly is assembled to the filter assembly lens holder, and the filter assembly lens holder is assembled on a top side of the molding base.

(48) In the above photosensitive assembly of the camera module, further including a protective frame provided on the photosensitive chip, and the molding base is integrally molded on the circuit board, the photosensitive chip and the protective frame.

(49) According to still another aspect of the present application, the present application provides a photosensitive assembly jointed panel of a camera module, comprising: one or more rows of circuit boards, each row of circuit boards including one or more circuit boards arranged side by side, each of the circuit boards including a combined rigid region and flexible region; one or more rows of photosensitive chips; and one or more one-piece molding bases, each of which is integrally molded on a row of the circuit boards and a row of the photosensitive chips and forms a light window for providing a light path for each of the photosensitive chips; wherein a distance between an outer edge and an inner edge of a portion of the one-piece molding base corresponding to a first end side of the one-piece molding base adjacent to the flexible region is a; a distance between an outer edge and an inner edge of the one-piece molding base corresponding to an opposite second end side of the one-piece molding base away from the flexible region is c, wherein 0.2 mm≤a≤1 mm, 0.2 mm≤c≤1.5a.

(50) In the above photosensitive assembly jointed panel of the camera module, the photosensitive assembly jointed panel includes a plurality of rows of the circuit boards, a plurality of rows of the photosensitive chips, and a plurality of the one-piece molding bases, wherein in two adjacent rows of the circuit boards, the second end side of the one-piece molding base on one row of the circuit boards away from the flexible region faces towards the first end side of the one-piece molding bases adjacent to the flexible region on another row of the circuit boards.

(51) In the above photosensitive assembly jointed panel of the camera module, there is 0.7a≤c≤1.3a. In the above photosensitive assembly jointed panel of the camera module, the rigid region of each of the circuit boards has a pressing distance W for a pressing block of the first molding die in the molding process to facilitate pressing on the first end side adjacent to the flexible region, and the pressing distance W has a value ranging from 0.1˜1 mm.

(52) In the above photosensitive assembly jointed panel of the camera module, the rigid regions of a row of the circuit boards are integrally molded to form an integral rigid region jointed panel.

(53) According to still another aspect of the present application, the present application provides a photosensitive assembly jointed panel of a camera module, comprising: a plurality of rows of circuit boards, each row of circuit boards including one or more circuit boards arranged side by side, each of the circuit boards including a combined rigid region and flexible region; a plurality of rows of photosensitive chip; and one or more one-piece molding bases, each of which is integrally molded on two rows of adjacent circuit boards and two rows of adjacent photosensitive chips and forms a light window for providing a light path for each of the photosensitive chips; and the two adjacent rows of the circuit boards are arranged such that the flexible regions thereof are far from each other and the rigid regions thereof are adjacent to each other, thereby each of the one-piece molding bases has two end sides adjacent to the flexible region; wherein a distance between an outer edge and an inner edge of a portion of the one-piece molding base corresponding to each end side of the one-piece molding base adjacent to the flexible region is a; the one-piece molding base

extends to be located between the two adjacent rows of the photosensitive chips and a distance between two inner edges thereof is c, wherein 0.2 mm≤a≤1 mm, 0.2 mm≤c≤1.5a.
(54) In the above photosensitive assembly jointed panel of the camera module, the rigid region of each of the circuit boards has a pressing distance W for a pressing block of the first molding die in the molding process to facilitate pressing on the first end side adjacent to the flexible region, and the pressing distance W has a value ranging from 0.1˜1 mm.
(55) In the above photosensitive assembly jointed panel of the camera module, the rigid regions of the two adjacent rows of the circuit boards are integrally molded to form an integral rigid region jointed panel.
(56) According to still another aspect of the present application, the present application provides a camera module, comprising: a lens; a circuit board, including a combined rigid region and flexible region; a photosensitive chip; and a molding base, wherein the molding base is integrally molded on the circuit board and the photosensitive chip and forms a light window for providing a light path for the photosensitive chip, wherein the lens is located on a photosensitive path of the photosensitive chip; wherein a distance between an outer edge and an inner edge of a portion of the molding base corresponding to a first end side of the molding base adjacent to the flexible region is a; a distance between an outer edge and an inner edge of a portion of the molding base corresponding to an opposite second end side of the molding base away from the flexible region is c, wherein 0.2 mm≤a≤1 mm, 0.2 mm≤c≤1.5a.
(57) In the above camera module, there is 0.7a≤c≤1.3a.
(58) In the above camera module, the circuit board further includes a plurality of electronic components, wherein the plurality of electronic components are provided at at least one wing side on both sides of the photosensitive assembly on the rigid region except for both end sides being adjacent to the flexible region and away from the flexible region, wherein the molding base integrally embeds the electronic component.
(59) In the above camera module, the molding base integrally molded on the circuit board and the photosensitive chip is cut from a one-piece molding base made in a jointed panel molding process, and the opposite second end side of the molding base away from the flexible region corresponds to a cutting side.
(60) In the above camera module, a distance between an outer edge and an inner edge of the cutting side of the molding base corresponding to the opposite second end side of the molding base away from the flexible region is c, where 0.2 mm≤c≤1.5a-0.2 mm.
(61) In the above camera module, further including a filter assembly and a filter assembly lens holder, wherein the filter assembly is assembled to the filter assembly lens holder, the filter assembly lens holder is assembled on the top side of the molding base.
(62) In the above camera module, the lens is assembled on the molding base; or the lens is assembled on the filter assembly lens holder; or the lens is assembled on the molding base and the filter assembly lens holder.
(63) In the above camera module, further including a driver, the lens is assembled on the driver, the driver is assembled on the molding base, or the driver is assembled on the filter assembly lens holder; or the driver is assembled on the molding base and the filter assembly lens holder.
(64) In the above camera module, further including a fixed lens barrel, the lens is assembled on the fixed lens barrel, and the fixed lens barrel is assembled on the molding base; or the fixed lens barrel is assembled on the filter assembly lens holder; or the fixed lens barrel is assembled on the molding base and the filter assembly lens holder.
(65) According to still another aspect of the present application, the present application provides a molding die to produce a photosensitive assembly jointed panel for a camera module, comprising a first die and a second die suitable for being separated from each other and being in close contact with each other, wherein the first die and the second die form a molding cavity when they are in close contact with each other, and the molding die is configured with a light window molding

portion and a base jointed panel molding guide groove formed around the light window molding portion in the molding cavity, and the molding cavity is suitable for fixing a circuit board jointed panel therein, wherein the circuit board jointed panel includes one or more rows of circuit boards, each row of circuit boards includes one or more circuit boards arranged side by side, each of the circuit boards includes a combined rigid region and flexible region, and each of the circuit boards is operatively connected with a photosensitive chip, and the base jointed panel molding guide groove is adapted to be filled with a molding material to form a one-piece molding base at a position corresponding to the base jointed panel molding guide groove, wherein the one-piece molding base is integrally molded in corresponding each row of the circuit boards and each row of the photosensitive chips to form the photosensitive assembly jointed panel and form a light window for providing a light path for each of the photosensitive chips at a position corresponding to the light window molding portion, wherein the base jointed panel molding guide groove has a first diversion groove corresponding to a first end side of the one-piece molding base adjacent to the flexible region and a second diversion groove corresponding to the one-piece molding base away from the flexible region, and a plurality of filling grooves extending between the first diversion groove and the second diversion groove, wherein each of the light window molding portions is located between the two adjacent filling grooves, wherein a width of a bottom end of the first diversion groove is a, and a width of a bottom end of the second diversion groove is c, where 0.2 mm≤a≤1 mm, 0.2 mm≤c≤1.5a.

(66) According to still another aspect of the present application, the present application provides a molding die to produce a photosensitive assembly jointed panel for a camera module, comprising a first die and a second die suitable for being separated from each other and being in close contact with each other, wherein the first die and the second die form a molding cavity when they are in close contact with each other, and the molding die is configured with a light window molding portion and a base jointed panel molding guide groove formed around the light window molding portion in the molding cavity, and the molding cavity is suitable for fixing a circuit board jointed panel therein, wherein the circuit board jointed panel includes a plurality of rows of circuit boards, each row of circuit boards includes one or more circuit boards arranged side by side, each of the circuit boards includes a combined rigid region and flexible region, and each of the circuit boards is operatively connected with a photosensitive chip, wherein the base jointed panel molding guide groove is adapted to be filled with a molding material to form a one-piece molding base at a position corresponding to the base jointed panel molding guide groove, wherein the one-piece molding base is integrally molded on the two adjacent rows of the circuit boards and the two adjacent rows of the photosensitive chips to form the photosensitive assembly jointed panel and from a light window for providing a light path for each of the photosensitive chips at a position corresponding to the light window molding portion, wherein the two adjacent rows of the circuit boards are arranged such as their flexible regions are away from each other and their rigid regions are adjacent to each other, wherein the base jointed panel molding guide groove has two first diversion grooves corresponding to two end sides of the one-piece molding base adjacent to the flexible region and a second diversion groove corresponding to a region between the two adjacent rows of the photosensitive chips, and a plurality of filling grooves extending between the first diversion groove and the second diversion groove, wherein each of the light window molding portions is located between the two adjacent filling grooves, wherein a width of a bottom end of the first diversion groove is a, and a width of a bottom end of the second diversion groove is c, where 0.2 mm≤a≤1 mm, 0.2 mm≤c≤1.5a.

(67) According to still another aspect of the present application, the present application provides a method for manufacturing a photosensitive assembly of a camera module, comprising the following steps: (1) fixing a circuit board jointed panel to a second die of a molding die, wherein the circuit board jointed panel includes one or more rows of circuit boards, and each row of circuit boards includes one or more circuit boards arranged side by side, each of the circuit boards

includes a combined rigid region and flexible region, and each of the circuit boards is operatively connected with a photosensitive chip; (2) clamping the second die and a first die and filling molten molding material in a base jointed panel molding guide groove in the molding die, wherein a position corresponding to a light window molding portion is prevented from being filled with the molding material; and (3) curing the molding material in the base jointed panel molding guide groove to form a one-piece molding base at a position corresponding to the base jointed panel molding guide groove, wherein the one-piece molding base is integrally molded on corresponding each row of the circuit boards and each row of the photosensitive chips to form a photosensitive assembly jointed panel and form a light window for providing a light path for each of the photosensitive chips at a position corresponding to the light window molding portion, wherein the base jointed panel molding guide groove has a first diversion groove corresponding to a first end side of the one-piece molding base adjacent to the flexible region and a second diversion groove corresponding to the one-piece molding base away from the flexible region, and a plurality of filling grooves extending between the first diversion groove and the second diversion groove, wherein each of the light window molding portions is located between two adjacent filling grooves, wherein a width of a bottom end of the first diversion groove is a, and a width of a bottom end of the second diversion groove is d, wherein 0.2 mm≤a≤1 mm, d>1.5a, and a flow retarding part is provide on the rigid region corresponding to the second end side of the one-piece molding base away from the flexible region, and the flow retarding part is embedded by the one-piece molding base.

(68) In the above method for manufacturing the photosensitive assembly of the camera module, further including a step of cutting the photosensitive assembly jointed panel to obtain a plurality of photosensitive assemblies, wherein each of the photosensitive assemblies includes the circuit board, the photosensitive chip, and the molding base, wherein the molding base is integrally molded on the circuit board and the photosensitive chip and forms the light window for providing a light path for the photosensitive chip.

(69) In the above method for manufacturing the photosensitive assembly of the camera module, further including a step of removing a part of the one-piece molding base for embedding the flow retarding part.

(70) According to still another aspect of the present application, the present application provides a method for manufacturing a photosensitive assembly of a camera module, comprising the following steps: (a) fixing a circuit board jointed panel to a second die of a molding die, wherein the circuit board jointed panel includes a plurality of rows of circuit boards, and each row of circuit boards includes one or more circuit boards arranged side by side, each of the circuit boards includes a combined rigid region and flexible region, and each of the circuit boards is operatively connected with a photosensitive chip; (b) clamping the second die and a first die and filling molten molding material in a base jointed panel molding guide groove in the molding die, wherein a position corresponding to a light window molding portion is prevented from being filled with the molding material; and (c) curing the molding material in the base jointed panel molding guide groove to form a one-piece molding base at a position corresponding to the base jointed panel molding guide groove, wherein the one-piece molding base is integrally molded on two adjacent rows of the circuit boards and two adjacent rows of the photosensitive chips to form a photosensitive assembly jointed panel and form a light window for providing a light path for each of the photosensitive chips at a position corresponding to the light window molding portion, wherein the two adjacent rows of the circuit boards are arranged such that their flexible regions are away from each other and their rigid regions are adjacent to each other, wherein the base jointed panel molding guide groove has two first diversion grooves corresponding to two end sides of the one-piece molding base adjacent to the flexible region and a second guide groove corresponding to a region between the two adjacent rows of photosensitive chips, and a plurality of filling grooves extending between two first diversion grooves and the second diversion groove, wherein each of

the light window molding portions is located between two adjacent filling grooves, wherein a width of a bottom end of the first diversion groove is a, and a width of a bottom end of the second diversion groove is d, wherein 0.2 mm≤a≤1 mm, d>1.5a, and a flow retarding part is provided in second diversion groove on a hard plate.
(71) In the above method for manufacturing the photosensitive assembly of the camera module, further including a step of cutting the photosensitive assembly jointed panel to obtain a plurality of photosensitive assemblies, wherein each of the photosensitive assemblies includes the circuit board, the photosensitive chip, and the molding base, wherein the molding base is integrally molded on the circuit board and the photosensitive chip and forms the light window that provides a light path for the photosensitive chip.
(72) In the above method for manufacturing the photosensitive assembly of the camera module, further including a steps of removing a part of the one-piece molding base between the two adjacent rows of the photosensitive chips and removing a part for embedding the flow retarding part.
(73) According to another aspect of the present application, the present application also provides an electronic device including one or more of the camera modules described above. The electronic device includes, but is not limited to, a mobile phone, a computer, a television, a smart wearable device, a vehicle, a camera, and a monitoring device.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) FIG. **1**A is a schematic structural view of a molding die for obtaining a photosensitive assembly packaged by the conventional integrated packaging process.
(2) FIG. **1**B is a schematic view of a molding process of forming an integrated packaging assembly by the conventional integrated package process.
(3) FIG. **1**C is an enlarged structure view illustrating the forward flow of the packaging material along with two flow passages in the conventional integrated packaging process.
(4) FIG. **1**D is an enlarged structure view illustrating that the packaging material is partially not filled with in the conventional integrated packaging process.
(5) FIG. **1**E is an enlarged structure view of damage to lead due to friction between leads in the conventional integrated packaging process.
(6) FIG. **1**F is a schematic view of a viscosity change tendency of a molding material during curing time.
(7) FIG. **2** is a schematic block view of a manufacturing equipment of a photosensitive assembly jointed panel for a camera module according to the first preferred embodiment of the present application.
(8) FIG. **3**A is a schematic structural view of a molding die of the manufacturing equipment for the photosensitive assembly jointed panel of the camera module according to the above mentioned first preferred embodiment of the present application.
(9) FIG. **3**B is an enlarged structural schematic view of a partial region A of a first die of the molding die of the manufacturing equipment of the photosensitive assembly jointed panel of the camera module according to the above mentioned first preferred embodiment of the present application.
(10) FIG. **4** is a schematic structural view of the photosensitive assembly jointed panel of the camera module according to the first preferred embodiment of the present application.
(11) FIG. **5**A is an enlarged structural schematic view of the photosensitive assembly of the camera module according to the first preferred embodiment of the present application.
(12) FIG. **5**B is an enlarged structural schematic top view of the photosensitive assembly of the

camera module according to the first preferred embodiment of the present application.
(13) FIG. **6**A is a cross-sectional view of the photosensitive module of the camera module according to the above mentioned first preferred embodiment of the present application, taken along the line C-C in FIG. **5**A.
(14) FIG. **6**B is a cross-sectional view of a second end side of the photosensitive module of the camera module according to the above mentioned first preferred embodiment of the present application after being further cut.
(15) FIG. **7**A is a cross-sectional view illustrating when a molten molding material is pushed into a base jointed panel molding guide groove in the molding die of the photosensitive assembly jointed panel according to the above mentioned first preferred embodiment of the present application, wherein the cross-sectional view is a cross-sectional view corresponding to the direction of the A-A line illustrated in FIG. **4**.
(16) FIG. **7**B is a partially enlarged schematic view at B in FIG. **7**A.
(17) FIG. **8** is a cross-sectional view of a molten molding material being filled in a base jointed panel molding guide groove in the molding die of the photosensitive assembly jointed panel according to the above mentioned first preferred embodiment of the present application, wherein the cross-sectional view is a cross-sectional view corresponding to the direction of the A-A line illustrated in FIG. **4**.
(18) FIG. **9** is a cross-sectional view of a molten molding material being filled in a base jointed panel molding guide groove in the molding die of the photosensitive assembly jointed panel according to the above mentioned first preferred embodiment of the present application, wherein the cross-sectional view is a cross-sectional view corresponding to the direction of the B-B line illustrated in FIG. **4**.
(19) FIG. **10** is a cross-sectional view of performing a demold step to form a one-piece molding base in the molding die of the photosensitive assembly jointed panel according to the above mentioned first preferred embodiment of the present application, wherein the cross-sectional view is a cross-sectional view corresponding to the direction of the A-A line illustrated in FIG. **4**.
(20) FIG. **11** is a schematic view illustrating a three-dimensional structure of the camera module according to the above mentioned first preferred embodiment of the present application.
(21) FIG. **12** is a schematic exploded view illustrating the camera module according to the above mentioned first preferred embodiment of the present application.
(22) FIG. **13**A is a cross-sectional view illustrating the camera module according to the above mentioned first preferred embodiment of the present application, taken along line D-D in FIG. **12**.
(23) FIG. **13**B is a cross-sectional view illustrating the camera module according to the above mentioned first preferred embodiment of the present application, taken along line E-E in FIG. **12**.
(24) FIG. **14** is a cross-sectional view illustrating a variant embodiment of the camera module according to the above mentioned first preferred embodiment of the present application.
(25) FIG. **15** is a cross-sectional view illustrating a camera module according to another variant embodiment of the camera module of the above mentioned first preferred embodiment of the present application.
(26) FIG. **16** is a cross-sectional view of a camera module according to another variant embodiment of the camera module of the above mentioned first preferred embodiment of the present application.
(27) FIG. **17**A is a schematic structural view of a molding die of a manufacturing equipment of a photosensitive assembly jointed panel of a camera module according to a second preferred embodiment of the present application.
(28) FIG. **17**B is an enlarged schematic structural view of a partial region C of a first die of the molding die of the manufacturing equipment of the photosensitive assembly jointed panel of the camera module according to the above mentioned second preferred embodiment of the present application.

(29) FIG. **18** is a schematic structural view of the photosensitive assembly jointed panel of the camera module according to the above mentioned second preferred embodiment of the present application.
(30) FIG. **19**A is an enlarged structural schematic view of D position of the photosensitive assembly jointed panel of the camera module according to the above mentioned second preferred embodiment of the present application.
(31) FIG. **19**B is an enlarged structural schematic top view of two adjacent photosensitive assemblies of the photosensitive assembly jointed panel of the camera module according to the above mentioned second preferred embodiment of the present application.
(32) FIG. **20**A is a cross-sectional view of the photosensitive assembly jointed panel of the camera module according to the above mentioned second preferred embodiment of the present application, taken along line H-H in FIG. **19**A.
(33) FIG. **20**B is a schematic structural view of two photosensitive assemblies obtained by cutting the photosensitive assembly jointed panel of the camera module according to the above mentioned second preferred embodiment of the present application.
(34) FIG. **21**A is a cross-sectional view illustrating when a molten molding material is pushed into a base jointed panel molding guide groove in the molding die of the photosensitive assembly jointed panel according to the above mentioned second preferred embodiment of the present application, wherein the cross-sectional view is a cross-sectional view corresponding to the direction of the F-F line illustrated in FIG. **18**.
(35) FIG. **21**B is a partially enlarged schematic view at E in FIG. **21**A.
(36) FIG. **22** is a cross-sectional view illustrating a molten molding material being filled in a base jointed panel molding guide groove in the molding die of the photosensitive assembly jointed panel according to the above mentioned second preferred embodiment of the present application, wherein the cross-sectional view is a cross-sectional view corresponding to the direction of the F-F line illustrated in FIG. **18**.
(37) FIG. **23** is a cross-sectional view illustrating a molten molding material being filled in a base jointed panel molding guide groove in the molding die of the photosensitive assembly jointed panel according to the above mentioned second preferred embodiment of the present application, wherein the cross-sectional view is a cross-sectional view corresponding to the direction of the G-G line illustrated in FIG. **18**.
(38) FIG. **24** is a cross-sectional view of performing a demold step to form a one-piece molding base in the molding die of the photosensitive assembly jointed panel according to the above mentioned second preferred embodiment of the present application, wherein the cross-sectional view is a cross-sectional view corresponding to the direction of the F-F line illustrated in FIG. **18**.
(39) FIG. **25**A to **25**C are schematic cross-sectional views of a photosensitive assembly jointed panel according to a variant embodiment of the above mentioned first and second preferred embodiments of the present application, and enlarged structural views of the photosensitive assembly obtained by cutting the photosensitive assembly jointed panel.
(40) FIG. **26**A is a schematic structural view illustrating a photosensitive assembly jointed panel according to another variant embodiment of the above mentioned second preferred embodiment of the present application.
(41) FIG. **26**B is a schematic enlarged structure view of a photosensitive assembly according to another variant embodiment of the above mentioned second preferred embodiment of the present application.
(42) FIG. **27** is a cross-sectional view illustrating a photosensitive assembly according to another variant embodiment of the above mentioned second preferred embodiment of the present application, taken along the line I-I in FIG. **26**B.
(43) FIG. **28**A is a schematic structural view of a molding die of a manufacturing equipment of a photosensitive assembly jointed panel of a camera module according to the third preferred

embodiment of the present application.
(44) FIG. **28**B is an enlarged structural schematic view of a partial region F of a first die of the molding die of the manufacturing equipment of the photosensitive assembly jointed panel of the camera module according to the above mentioned third preferred embodiment of the present application.
(45) FIG. **29** is a schematic structural view of the photosensitive assembly jointed panel of the camera module according to the above mentioned third preferred embodiment of the present application.
(46) FIG. **30** is a cross-sectional view of the photosensitive assembly jointed panel of the camera module according to the above mentioned third preferred embodiment of the present application, taken along line J-J in FIG. **29**.
(47) FIG. **31** is an enlarged structural cross-sectional view of the photosensitive assembly of the camera module according to the above mentioned third preferred embodiment of the present application after an embedded flow retarding part is removed.
(48) FIG. **32** is a cross-sectional view illustrating when a molten molding material is filled in a base jointed panel molding guide groove in the molding die of the photosensitive assembly jointed panel according to the above mentioned third preferred embodiment of the present application, wherein the cross-sectional view is a cross-sectional view corresponding to the direction of the J-J line illustrated in FIG. **29**.
(49) FIG. **33**A is a schematic structural view of a molding die of a manufacturing equipment for a photosensitive assembly jointed panel of a camera module according to the fourth preferred embodiment of the present application.
(50) FIG. **33**B is an enlarged structure schematic view of a partial region G of a first die of the molding die of the manufacturing equipment for the photosensitive assembly jointed panel of the camera module according to the above mentioned fourth preferred embodiment of the present application.
(51) FIG. **34** is a structural schematic view of the photosensitive assembly jointed panel of the camera module according to the above mentioned fourth preferred embodiment of the present application.
(52) FIG. **35** is a cross-sectional view of the photosensitive assembly jointed panel of the camera module according to the above mentioned fourth preferred embodiment of the present application, taken along line K-K in FIG. **34**.
(53) FIG. **36** is an enlarged structural cross-sectional view of the photosensitive assembly of the camera module according to the above mentioned fourth preferred embodiment of the present application after an embedded flow retarding part is removed.
(54) FIG. **37** is a cross-sectional view illustrating when a molten molding material is filled in a base jointed panel molding guide groove in the molding die of the photosensitive assembly jointed panel according to the above mentioned fourth preferred embodiment of the present application, wherein the cross-sectional view is a cross-sectional view corresponding to the direction of the K-K line illustrated in FIG. **34**.
(55) FIG. **38** is a structural schematic view of the above-mentioned camera module applied to an intelligent electronic device according to the present application.

DETAIL DESCRIPTION OF THE INVENTION

(56) The following description is used to disclose the present application so that those skilled in the art can implement the present application. The preferred embodiments in the following description are merely examples, and those skilled in the art can think of other obvious variations. The basic principles of the present application defined in the following description can be applied to other embodiments, modifications, improvements, equivalents, and other technical solutions without departing from the spirit and scope of the present application.
(57) Those skilled in the art should understand that, in the disclosure of the present application, The

orientation or positional relationship of the indications of the terms “longitudinal”, “transverse”, “upper”, “lower”, “front”, “back”, “left”, “right”, “upright”, “horizontal”, “top”, “bottom”, “inside”, “outside”, etc. is based on the orientation or positional relationship shown in the drawings, which is merely for the convenience of describing the present application and simplifying the description rather than indicating or implying that the device or component referred to must have a particular orientation, be constructed and operated in a particular orientation. Therefore, the above terms are not to be construed as limiting the present invention.

(58) It can be understood that the term “a” should be understood as “at least one” or “one or more”, that is, in one example, the number of a component can be one, while in other examples, the number of the component may be plural, and the term “a” cannot be understood as a limitation on the number.

(59) As shown in FIG. **2** to FIG. **14**, a camera module **100** and a photosensitive assembly **10** and a manufacturing method thereof according to a first preferred embodiment of the present application are shown. The camera module **100** can be applied to various electronic devices **300**. The electronic device **300** includes a device main body **301** and one or more camera modules **100** mounted in the device main body **301**, as shown in FIG. **38**. The electronic device **30** is exemplified but not limited to a smart phone, a wearable device, a computer device, a television, a vehicle, a camera, a monitoring device, etc., and the camera module cooperates with the electronic device to implement image acquisition and reproduction of the target object.

(60) More specifically, the figure illustrates a photosensitive assembly **10** of the camera module **100** and its manufacturing equipment **200**. The photosensitive assembly **10** includes a circuit board **11**, a molding base **12** and a photosensitive chip **13**. The molding base **12** is integrally molded on the circuit board **11** and the photosensitive chip **13** and forms a light window **122** for providing a light path for the photosensitive chip **13**. Wherein, the molding base **12** of the present application is integrally molded on the circuit board **11** and the photosensitive chip **13** by the manufacturing equipment **200** through a molding process, more specifically, a transfer molding process, so that the molding base **12** can replace the lens holder or the bracket of the conventional camera module, and does not need to attach the lens holder or the bracket to the circuit board **11** by glue in a conventional packaging process.

(61) Further, referring to FIGS. **2-4** and FIGS. **7**A to **10**, the present application manufactures a photosensitive assembly jointed panel **1000** through the manufacturing equipment **200**, that is, the photosensitive assembly jointed panel **1000** having a plurality of photosensitive assemblies **10** is manufactured by a jointed panel process. The photosensitive assembly jointed panel **1000** includes a circuit board **1100** and one or more one-piece molding bases **1200**. The circuit board jointed panel **1100** includes a plurality of rows of circuit boards, such as the four-row circuit boards illustrated in FIG. **4**, and each row of the circuit boards includes a plurality of circuit boards **11**, such as **2-12** of the circuit boards **11**, six of the circuit boards shown the FIG. **11**, and each of the circuit boards **11** is operatively connected to a photosensitive chip **13**. Each of the one-piece molding bases **1200** is formed on a row of the circuit boards and is integrally molded on at least a part of a non-photosensitive region **132** of each of the photosensitive chips **13** of the row of the photosensitive chips **13** to expose a photosensitive region **131** of the photosensitive assembly **13**. Each of the one-piece molding bases **1200** has a plurality of light windows **122**, and the positions of each of the light windows **122** correspond to each of the photosensitive chips **13** for providing a light path for the corresponding photosensitive chips **13**.

(62) The manufacturing equipment **200** of the photosensitive assembly jointed panel **1000** of the camera module **100** includes a molding die **210**, a molding material feeding mechanism **220**, a die fixing device **230**, a temperature control device **250**, and a controller **260**. The molding material feeding mechanism **220** is used to provide a molding material **14** to a base jointed panel molding guide groove **215**. The die fixing device **230** is used to control the die opening and clamping of the molding die **210**. The temperature control device **250** is used to heat the thermosetting molding

material **14**.

(63) The controller **260** is used to automatically control the operation of the molding material feeding mechanism **220**, the die fixing device **230**, and the temperature control device **250** in the molding process.

(64) The molding die **210** includes a first die **211** and a second die **212** that can be opened and clamped under the action of the die fixing device **230**, that is, the die fixing device **230** can operate the first die **211** and the second die **212** being separated and closely contacted to form a molding cavity **213**. During die clamping, the circuit board jointed panel **1100** is fixed in the molding cavity **213**, and the fluid-like molding material **14** enters the molding cavity **213** so as to be integrally molded on each row of the circuit boards **11** and the corresponding each row of the photosensitive chips **13**, and after curing, it forms the one-piece molding base **1200** integrally molded on each row of the circuit boards **11** and on each row of the photosensitive chips **13**.

(65) More specifically, the molding module **210** further has one or more base jointed panel molding guide grooves **215** and a plurality of light window molding portions **214** located in the base jointed panel molding guide grooves **215**. When the first and second dies **211** and **212** are clamped, the light window molding portion **214** and the base jointed panel molding guide groove **215** extend in the molding cavity **213**, and the fluid-like molding material **14** is filled into the base jointed panel molding guide groove **215**, and the position corresponding to the light window molding portion **214** cannot be filled with the fluid-like molding material **14**, therefore, at a position corresponding to the base jointed panel molding guide groove **215**, the fluid-like molding material **14** can be formed into the one-piece molding base **1200** after curing, which includes a ring-shaped molding main body **121** corresponding to the molding base **12** of each of the photosensitive assemblies **10**, and the light window **122** of the molding base **12** is formed at a position corresponding to the light window molding portion **214**. The molding material **14** may be selected from, but not limited to, nylon, LCP (Liquid Crystal Polymer), PP (Polypropylene), epoxy resin, and the like.

(66) The first die **211** and the second die **212** may be two dies capable of generating relative movement, such as one of the two molds is fixed and the other is movable; or both molds are movable. The present application is not limited in this aspect. In the example of this embodiment of the present application, the first die **211** is specifically implemented as a fixed upper die, and the second die **212** is implemented as a movable lower die. The fixed upper die and the movable lower die are arranged coaxially, for example, the movable lower die can slide up along with a plurality of positioning axes, and can form the tightly closed molding cavity **213** when clamping die with the fixed upper die.

(67) The second die **212**, that is, the lower die may have a circuit board positioning groove **2121**, which may be in the shape of a groove or formed by a positioning post for mounting and fixing the circuit board **11**, and the light window molding portion **214** and the base jointed panel molding guide groove **215** may be formed in the first die **211**, that is, formed in the upper die, and when the first die **211** and the second die **212** are clamped, it forms the molding cavity **213**. And, the fluid-like molding material **14** is injected into the base jointed panel molding guide grooves **215** on the top side of the circuit board jointed panel **1100**, so that the one-piece molding base **1200** is formed on the top side of each row of the circuit boards **11** and each row of the photosensitive chips **13**.

(68) It can be understood that the circuit board positioning groove **2121** may also be provided in the first die **211**, that is, the upper die, for mounting and fixing the circuit board jointed panel **1100**, and the light window molding portion **214** and the base jointed panel molding guide groove **215** may be formed in the second die **211**, and when the first die **211** and the second die **212** are clamped, the molding cavity **213** is formed. The circuit board jointed panel **1100** may be arranged face-to-face in the upper die, and the fluid-like molding material **14** is injected into the base jointed panel molding guide groove **215** on the bottom side of the circuit board jointed panel **1100** that is inverted, so that the one-piece molding base **1200** is formed on the bottom side of the circuit board jointed panel **1100** that is inverted.

(69) More specifically, when the first die **211** and the second die **212** are clamped and a molding step is performed, the light window molding portion **214** is superimposed on the top surface of the photosensitive chip **13** and closely adhered, so that the fluid-like molding material **14** is prevented from entering the photosensitive region **131** on the top surface of the photosensitive chip **13** on the circuit board **11**, so that the light window **122** of the one-piece molding base **1200** can be finally formed at a position corresponding to the light window molding portion **214**. It can be understood that the light window molding portion **214** may be a solid structure or a structure having a groove shape inside as shown in the figure.
(70) It can be understood that the molding surface of the first die **211** forming the base jointed panel molding guide groove **215** can be configured as a flat surface and is on the same plane. In this way, when the molding base **12** is cured and molded, the top surface of the molding base **12** is relatively flat, so as to provide flat mounting conditions for optical components such as drivers, lenses, and fixed lens barrels above the photosensitive assembly **10** of the camera module **100** to reduce the tilt error of the camera module **100** after assembly.
(71) It is worth mentioning that the base jointed panel molding guide groove **215** and the light window molding portion **214** can be integrally molded in the first die **211**. Alternatively, the first die **211** further includes a detachable molding structure, and the molding structure is formed with the base jointed panel molding guide groove **215** and the light window molding portion **214**. In this way, according to the shape and dimension requirements of the photosensitive assembly **10** such as the diameter and thickness of the molding base, the base jointed panel molding guide grooves **215** and the light window **214** of different shapes and dimensions can be designed. In this way, only the different molding structures need to be replaced, then the manufacturing equipment can be adapted to be applied to the photosensitive assembly **10** with different specifications. It can be understood that the second die **212** may also include accordingly a detachable fixing block to provide the grooves of different shapes and dimensions, so as to facilitate the replacement of the circuit board **11** adapted to different shapes and dimensions.
(72) It can be understood that the molding material **14** is a thermosetting material. The molding material **14** becomes a fluid state by heating and melting the thermosetting material in a solid state. During the molding process, the thermosetting molding material **14** is cured through a further heating process, and can no longer be melted after curing, thereby forming the one-piece molding base **1200**. It can be understood that, in the molding process of the present application, the molding material **14** may be a block shape, a granular shape, or a powder shape, which is changed to fluid in the molding die **210** after being heated, then cured to form the one-piece molding base **1200**.
(73) More specifically, each of the base jointed panel molding guide grooves **215** of the present application has a first diversion groove **2151** and a second diversion groove **2152** that are substantially parallel, and a plurality of filling grooves **2153** extends between the first diversion groove **2151** and the second diversion grooves **2152**, wherein the filling grooves **2153** are formed between two adjacent light window molding portions **214**, and as shown in the figure, the base jointed panel molding guide groove **215** has seven filling grooves **2153**, and six light window molding portions **214** are located between two adjacent filling grooves **2153**. The molding material **14** flows along with the first diversion groove **2151** and the second diversion groove **2152** from the feeding end **215**A to the terminal end **215**B, and between the first diversion groove **2151** and the second diversion groove **2152**, when the dimension and ratio of the first diversion groove **2151** and the second diversion groove **2152** are within a predetermined range, the molding material **14** can fill each of the filling grooves **2153**, so that the one-piece molding base **1200** is formed after the molding material **14** is cured.
(74) As shown in FIGS. **7**A to **10**, it is a schematic view of a manufacturing process of the photosensitive assembly jointed panel **1000** of the camera module **100** according to this preferred embodiment of the present application. As shown in FIG. **7**A, the molding die **210** is in a die clamping state, the circuit board jointed panel **1100** to be molded and the solid molding material **14**

are ready to be in place, the solid molding material **14** is heated, thereby when the molding material **14** is melted into a fluid state or a semi-solid or semi-fluid state, it is sent to the base jointed panel molding guide groove **215**, flows forward along with the first diversion groove **2151** and the second diversion groove **2152**, and fills the filling groove **2153** between two adjacent light window molding portions **214**.

(75) As shown in FIG. **8** and FIG. **9**, when the base molding plate forming guide groove **215** is completely filled with the fluid-like molding material **14**, the fluid-like molding material **14** is cured through a curing process to mold the one-piece molding base **1200** integrally molded in each row of the circuit boards **11** and each row of the photosensitive chips **13**.

(76) As shown in FIG. **10**, after the molding material **14** is cured to form the one-piece molding base **1200**, the demold process of the present application is performed, that is, the die fixing device **230** makes the first die **211** and the second die **212** be separated from each other, so that the light window molding portion **214** is separated from the one-piece molding base **1200**, so that the light window **122** corresponding to each of the photosensitive chips **13** is formed in the one-piece molding base **1200**.

(77) As shown in FIGS. **4** to **6**, the prepared photosensitive assembly jointed panel **1000** can be further cut to obtain a single photosensitive assembly **10**. Each of the photosensitive assemblies **10** includes at least one circuit board **11**, at least one photosensitive chip **13** and the molding base **12** integrally molded on the circuit board **11** and the photosensitive chip **13**. Each of the circuit boards **11** includes a combined rigid region **111** and flexible region **112**, that is to say, each of the circuit boards **11** can be implemented as a rigid-flex combined board in this example of the present application. Wherein, the molding base **12** integrally molds the rigid region **111** of the circuit board **11** and at least a part of the non-photosensitive region **132** of the photosensitive chip **13**, and forms the light window **122** that provides a light path for the photosensitive region **131** of the photosensitive chip **13**.

(78) It is worth mentioning that the manufacturing method of the photosensitive assembly jointed panel **1000** of the present application is suitable for manufacturing the photosensitive assembly **10** with a small dimension. In the molding process, the width of the bottom end of the first diversion groove **2151** is a, and the width of the bottom end of the second diversion groove **2152** is c. The width a corresponds to the width of the bottom end of the first flow guide **2151** adjacent to the position where the rigid region **111** and the flexible region **112** of the circuit board **11** are combined, and the width c corresponds to the width of the bottom end of the second diversion groove **2152** away from the other side of the flexible region **112**. When the widths a and c meet the following conditions, that is, $0.2\text{ mm} \leq a \leq 1\text{ mm}$ and $0.2\text{ mm} \leq c \leq 1.5a$, the fluid-like molding material **14** can flows forward along with the first diversion groove **2151** and the second diversion groove **2152** and fills the entire base jointed panel molding guide groove **215** with the molding material **14** before the molding material **14** is cured. It is worth mentioning that $0.7a \leq c \leq 1.3a$ is more preferred, for example, in some examples, $c=0.8a$ or $c=1a$ or $c=1.2a$.

(79) Correspondingly, the molding process of the present application yields the photosensitive assembly jointed panel **1000**, which includes: one or more rows of the circuit board **11**, one or more rows of the photosensitive chip **13**, and one or more of a one-piece molding base **1200**. Each row of the circuit boards **11** includes one or more of the circuit boards **11** arranged side by side, and each of the circuit boards **11** includes a combined rigid region **111** and flexible region **112**. Each of the one-piece molding bases **1200** is integrally molded in a row of the circuit boards **11** and a row of the photosensitive chips **13** and forms the light windows **122** that provide a light path for each of the photosensitive chips **13**. Wherein, a distance between the outer edge **1201** and the inner edge **1202** of the portion **1200**A of the one-piece molding base corresponding to the first end side of the one-piece molding base **1200** adjacent to the flexible region **112** is a; a distance between the outer edge **1203** and the inner edge **1204** of the portion **1200**B of the one-piece molding base corresponding to the opposite second end side of the one-piece molding base **1200** away from the

flexible region **112** is c, where 0.2 mm≤a≤1 mm, 0.2 mm≤c≤1.5a. Wherein, the first end side of the one-piece molding base **1200** corresponds to a combined side of the rigid region **111** and the flexible region **112** of the circuit board **11**, that is, near the proximal side of the flexible region **112**; the second end side of the one-piece molding base **1200** corresponds to the distal end side of the circuit board **11** away from the flexible region **112**.

(80) A single of the photosensitive assembly **10** can be obtained after the photosensitive assembly jointed panel **1000** is cut, wherein, in the cutting step, it can be cut on the two wing sides of the one-piece molded base **1200** except for the first end side and the second end side to obtain the molding base **12**, and the portion **1200**B of the molding base corresponding to the second end side is not cut, so that the photosensitive assembly **10** having the portion **1200**C of the one-piece molding base on a pair of opposite wing sides is obtained.

(81) As shown in FIG. **6**A, correspondingly, the photosensitive assembly **10** includes the circuit board **11**, the photosensitive chip **13**, and the molding base **12**. Wherein, the circuit board **11** includes a combined rigid region **111** and flexible region **112**. The molding base **12** is integrally molded on the circuit board **11** and the photosensitive chip **13** and forms the light window **122** that provides a light path for the photosensitive chip **13**. The circuit board **11** and the photosensitive chip **13** are connected through a series of connecting wires **15**. A distance between the outer edge **1201** and the inner edge **1202** of the portion **12**A of the molding base corresponding to the first end side of the molding base **12** adjacent to the flexible region **112** is a; a distance between the outer edge **1203** and the inner edge **1204** of the portion **12**B of the molding base portion corresponding to the opposite second end side of the molding base **12** away from the flexible region **112** is c, where 0.2 mm≤a≤1 mm, 0.2 mm≤c≤1.5a.

(82) As shown in FIG. **6**B, correspondingly, in order to further reduce the dimension of the photosensitive assembly **10**, at least a part of the photosensitive assembly **10** on the opposite second end side of the molding base **12** away from the flexible region **112** is suitable for being removed, such as cutting with a cutting tool, or being abraded, so that a distance between the outer edge **1205** and the inner edge **1204** of the remaining part of the molding base **12** after being cut is b, wherein 0.2 mm≤b≤1.5a-0.2 mm. Wherein, b is reduced relative to c by 0.2 mm, which corresponds to the minimum accuracy of the cutting tool, that is, the cutting dimension limit of the cutting tool enables the molding base **12** to be cut by a portion having a width of approximately 0.2 mm.

(83) Based on the consideration of the small-dimension photosensitive assembly **10**, a≤1 mm is selected. And according to actual production experience, when the dimensions of the widths a and c are relatively large, such as when both are larger than 1 mm, the flow velocity and flow rate in the two diversion grooves **2151** and **2152** are sufficient to fill the entire base jointed panel molding guide groove **215** before the molding material **14** is cured.

(84) It is found in actual production that when a<0.2 mm or c<0.2 mm, because the dimensions of the widths a and c are relatively small, the flow velocity and the flow rate of the mold material **14** in the corresponding first diversion groove **2151** and the second diversion groove **2152** are relatively small, so that the base plate molding guide groove **215** cannot be filled with the mold material **14** during the curing time of the molding material.

(85) Wherein, when c>1.5a, that is, when the value of c is greater than 1.5 times the value of a, such as c=1.6a, in actual production, a part of the base jointed panel molding guide groove **215** cannot be filled, resulting in a defective product. Therefore, in this example of the present application, 0.2 mm ≤a≤1 mm and 0.2 mm≤c≤1.5a. The above mentioned dimension range of the widths a and c enables the molding material **14** to fill the base jointed panel molding guide groove **215** in the molding process, thereby avoiding the occurrence of defective products of the photosensitive assembly.

(86) In this embodiment of the present application, when 0.2 mm≤a≤1 mm and 0.2 mm≤c≤1.5a, in the molding process, the molding material **14** is capable of forming one-piece molding base **1200**

on the circuit board jointed panel **1100**, and the one-piece molding base **1200** can form the light window **122** closed all around at the position corresponding to each of the photosensitive chips **13**, so that after cutting the formed one-piece photosensitive assembly jointed panel **1200**, a molding base **12** having the light window **122** is formed on each circuit board **11** and the corresponding photosensitive chip **13** to prevent a part of the molding base from forming an opening similar to that in FIG. **1**C to communicate the light window **122** to the outside of the molding base **12**.
(87) That is, the molding material **14** of the present application can flow forward from the feeding ends **215**A of the two diversion grooves **2151** and **2152** and fill the diversion grooves **2151** and **2152** and the filling groove **2153** of the entire base jointed panel molding guide groove **215**. The molding material **14** can flow along with the two diversion grooves **2151** and **2152** from the feeding end **215**A to the terminal end **215**B before curing. And before the viscosity of the molding material **14** reaches a high value and is cured, the molding material **14** can fill the base jointed panel molding guide groove **215**, thereby preventing the connecting wire **15** between the circuit board **11** and the photosensitive chip **13** from being damaged by the molding material **14** having a high viscosity and flowing forward. And the fluid in the two diversion grooves **2151** and **2152** flows forward at substantially the same step, and the two fluids basically meet in the filling groove **2153** to prevent the molding material **14** in one diversion groove from flowing to the other diversion groove and prevent the molding material **14** in the other diversion groove from flowing forward. Moreover, no turbulence or sinuous flow is generated, which causes the connecting wires **15** connecting the circuit board **11** and the photosensitive chip **13** to swing irregularly, resulting in deformation and damage.
(88) Accordingly, the molding material **14** of the present application can also select a material with a relatively high viscosity range, thereby avoiding that when a material with a small viscosity range is selected, the molding material **14** easily enters into the photosensitive region **131** of the photosensitive chip **13** to form a flash in the molding process.
(89) In addition, it is worth mentioning that, as shown in FIG. **7**B, in order to facilitate demold and pressing of the rigid region **111** of the circuit board **11**, the first die **211** further includes a plurality of pressing blocks **216**. The outer edge **1201** of the molding base **12** and the outer edge of the rigid region **111** of the circuit board **11** will form a pressing edge **1111** with a width of W, that is, in the molding process, it is suitable for the pressing block **216** is pressed on a region of the rigid region **111** of the circuit board **11**, for example, the pressing distance W may be 0.1˜1 mm, for example, in a specific example, the pressing distance W may be 0.2 mm. The pressing block **216** is further pressed over the flexible region **112** of each row of the circuit boards **11** to prevent the molding material **14** from flowing to the flexible region **112**. In addition, the rigid regions **111** of each row of the circuit boards **11** are integrally molded to form an integral rigid region jointed panel **110**, thereby facilitating the pressing of the first die to each row of the circuit boards **11**.
(90) Accordingly, the present application provides a method for manufacturing the photosensitive assembly **10** of the camera module **100**, which includes the following steps: fixing a circuit board jointed panel **1100** to a second die **212** of a molding die **210**, wherein the circuit board jointed panel **1100** includes one or more rows of circuit boards, and each row of circuit boards includes one or more circuit board **11** arranged side-by-side, each of the circuit boards **11** includes a combined rigid region **111** and flexible region **112**, and each of the circuit boards **111** is operatively connected to the photosensitive chip **13**; clamping the second die **212** and the first die **211** by the die fixing device **213** and filling the molten molding material **14** in a base jointed panel molding guide groove **215** in the molding die **210**, wherein the position corresponding to a light window molding portion **214** is prevented from being filled with the molding material **14**; curing the molding material **14** in the base jointed panel molding guide groove **215** to form a one-piece molding base **1200** at a position corresponding to the base jointed panel molding guide groove **215**, wherein the one-piece molding base **1200** is integrally molded on corresponding each row of the circuit boards **11** and each row of the photosensitive chips **13** to form a photosensitive assembly jointed panel **1000** and

form a light window **122** at a position corresponding to the light window molding portion **214** to provide a light path for each of the photosensitive chips **13**, wherein the base jointed panel molding guide groove **215** has a first diversion groove **2151** corresponding to a first end side of the one-piece molding base **1200** adjacent to the flexible region **112** and a second diversion groove **2152** corresponding to the one-piece molding base **1200** away from the flexible region **112**, and a filling grooves **2153** extending between the first diversion groove **2151** and the second diversion groove **2152** for filling the molding material **14** between two adjacent photosensitive chips **13** in each row of the photosensitive chips **13** and located between two adjacent light window molding portions **214**, wherein a width of the bottom end of the first diversion groove **2151** is a, and a width of the bottom end of the second diversion groove **2152** is c, wherein the width a corresponds to a distance between an outer edge **1201** and an inner edge **1202** of a portion **1200**A of the one-piece molding base of the first end side of the one-piece molding base **1200** adjacent to the flexible region **112**; wherein the width c corresponds to a distance between the outer edge **1203** and the inner edge **1204** of a portion **1200**B of the one-piece molding base of an opposite second end side of the one-piece molding base **1200** away from the flexible region **112**, wherein 0.2 mm≤a≤1 mm, 0.2 mm≤c≤1.5a, and preferably 0.7a≤c≤1.3a, such as c=0.8a or c=1a or c=1.2a, so that the width dimension of first diversion groove **2151** and the second diversion groove **2152** enables the molding material **14** to fill the base jointed panel molding guide groove **215** in the molding process of forming the one-piece molding base **1200**, and the molding material **14** can reach the terminal end **215**B of the first diversion groove **2151** and the second diversion groove **2152** from the feeding ends **215**A of the first diversion groove **2151** and the second diversion groove **2152**, respectively; cutting the photosensitive assembly jointed panel **1000** to obtain a plurality of the photosensitive assemblies **10**, wherein each of the photosensitive assemblies **10** includes the circuit board **11**, the photosensitive chip **13** and the molding base **12**, wherein, the molding base **12** is integrally molded on the circuit board **11** and the photosensitive chip **13** and forms the light window **122** that provides a light path for the photosensitive chip **13**.

(91) And, the method may further include a step of cutting a portion of the photosensitive assembly corresponding to the opposite second end side of the molding base **12** away from the flexible region **112**, that is, a part of the portion **12**B of the molding base and a part of the circuit board **11**, so that the molding base **12**B has a cutting surface **125** on an opposite second end side away from the flexible region **112**, and a distance between the outer edge **1203** and its inner edge **1204** of the remaining part of the molding base is b, where 0.2 mm≤b≤1.5a-0.2 mm.

(92) As shown in FIG. **5**A to FIG. **6**B, the circuit board **11** includes a plurality of electronic components **113** formed in the rigid region **111** such as mounted by SMT process. The electronic components **113** include, but are not limited to, resistors, capacitors, driving devices, etc. In this embodiment of the present application, the molding base **12** integrally covers the electronic component **113**, so as to prevent dust and debris from adhering to the electronic component **113** in similar conventional camera module, and further contaminating the photosensitive chip **13**, thereby affecting the imaging effect. In addition, preferably, the plurality of electronic components **113** are provided on a region except for a first end side **11**A and a second end side **11**B of the rigid region **111** of the circuit board **11** adjacent to the flexible area **112** and away from the flexible area **112**, at least one wing side **11**C on the rigid region **111** on both sides of the photosensitive assembly **11**, wherein the molding base **12** integrally embeds the electronic component **113**.

(93) That is, referring to FIGS. **8** and **9**, in the first diversion groove **2151** and the second diversion groove **2152**, the electronic component **113** is not included, and the electronic component **113** may be centrally provided in the filling groove **2153**, so that during the molding process, there will not be any block in the first diversion groove **2151** and the second diversion groove **2152**, so as not to affect the molding material **14** flows forward along with the first diversion groove **2151** and the second diversion groove **2152**, so that the molding material **14** flows from the feeding end **215**A to the terminal end **215**B as soon as possible.

(94) It can be understood that the connecting wires **15** may be provided on four sides of the photosensitive chip **13** or may be collectively provided on both wing sides **11**C of the rigid region **111** of the circuit board **11**, so that they are also concentrated in the filling groove **2153** during the molding process, thereby not affecting the forward flow of the molding material **14** along with the first diversion groove **2151** and the second diversion groove **2152**.
(95) As shown in FIGS. **11** to **14**, the camera module **100** to which the photosensitive assembly **10** of the present application is applied is illustrated. The camera module **100** includes a photosensitive assembly **10**, a lens **20** and a filter assembly **30**. The photosensitive assembly **10** includes the circuit board **11**, the molding base **12** and the photosensitive chip **13**. The lens **20** includes a structural member **21** and one or more lenses **22** accommodated in the structural member **21**. The filter assembly **30** includes a filter assembly lens holder **31** and a filter assembly **32**. The filter assembly lens holder **31** is assembled on the top side of the molding base **12**, and the lens **20** is directly assembled on the top side of the filter assembly lens holder **31** to form a fixed focus camera module. Wherein in this embodiment, the top side of the molding base **12** is a flat surface, the filter assembly lens holder **31** is assembled on the flat top surface of the molding base **12**, and the filter assembly **32** plays a role of filtering the light passing through the lens **20**, for example, it can be implemented as a filter for filtering infrared rays, which is located between the lens **20** and the photosensitive chip **13**.
(96) In this way, the light passing through the lens **30** can pass through the filter assembly **32** and reach the photosensitive chip **13** through the light window **122**, so that after the photoelectric conversion effect, the camera module **100** can provide an optical image.
(97) As shown in FIG. **13**A, in the photosensitive assembly **10** of the camera module **100**, a distance between an outer edge **1201** and an inner edge **1202** of a portion **12**A of the molding base corresponding to the first end side of the molding base **12** adjacent to the flexible region **112** is a; and a distance between an outer edge **1203** and an inner edge **1204** of a portion **12**B of the molding base corresponding to the opposite second end side of the molding base **12** away from the flexible region **112** is c, where 0.2 mm≤a≤1 mm, 0.2 mm≤c≤1.5a, so that the photosensitive assembly **10** can be obtained in a small dimension, thereby the dimension of the entire camera module **100** is further reduced. It can be understood that the opposite second end side of the molding base **12** away from the flexible region **112** can be further cut, so that the remaining part after the molding base **12** is cut has a cutting surface **125**, and the distance between an outer edge and an inner edge thereof is b, where 0.2 mm≤b≤1.5a-0.2 mm, as shown in FIG. **14**. In addition, as shown in FIG. **13**B, it can be seen that the electronic components **113** may be collectively provided on at least one of the two wing sides of the photosensitive assembly **10**, for example, may be concentrated on the two wing sides.
(98) It can be understood that, in another variant embodiment, the filter assembly lens holder **31** may not be provided, and the filter assembly **32** may be directly assembled on the molding base **12**, or the filter assembly **32** is assembled on the lens **20**, or the filter assembly **32** is assembled to a supporting member such as a driver or a fixed lens barrel of the lens **20**.
(99) As shown in FIG. **15**, the camera module **100** may include a supporting member **40**, which is a driver or a fixed lens barrel, in this figure, it is a driver such as a voice coil motor, a piezoelectric motor, etc., to form a moving focus camera module. The lens **20** is mounted on the driver. The molding base **12** has a groove **123** on the top side, which can be used to mount the filter assembly lens holder **31**, and the driver can be directly mounted on the top side of the molding base **12**. It can be understood that, in another variant embodiment, the supporting member **40** may also be mounted on the filter assembly lens holder **31**, or a part of the supporting member **40** may be mounted on the filter assembly lens holder **31**, and the other part thereof may be mounted on the molding base **12**.
(100) As shown in FIG. **16**, in this embodiment of the present application and in the drawings, the camera module **100** may include a supporting member **40**, which is a fixed lens barrel, and the lens

20 is mounted on the fixed lens barrel. The molding base **12** has a groove **123** on the top side, which can be used to mount the filter assembly lens holder **31**, and the fixed lens barrel is mounted on the top side of the molding base **12**.

(101) As shown in FIGS. **17**A to **24**, the photosensitive assembly **10** of the camera module **100** according to a second embodiment of the present application and a manufacturing process thereof are shown. In this embodiment, a photosensitive assembly jointed panel **1000** is also made by a jointed panel operation, and then cut to obtain the photosensitive assembly **10**. In the embodiments shown in FIGS. **2** to **16**, in a plurality of rows of circuit boards, the rigid regions **111** of one row of circuit boards are arranged adjacent to the flexible regions **112** of the other row of circuit boards. In this embodiment, the two adjacent rows of circuit boards can arrange the rigid regions **111** adjacently, and keep the corresponding flexible regions **112** away from each other. More preferably, the rigid regions **111** of the two adjacent rows of circuit boards are integrally molded, so that the middle of the two adjacent rows of circuit boards forms an integral rigid region.

(102) Accordingly, more specifically, the molding die **210** forms a molding cavity **213** when clamping, and provides a plurality of light window molding portions **214** and one or more base jointed panel molding guide grooves **215**, each of the base jointed panel molding guide groove **215** includes a first diversion groove **2151** arranged at two ends in a substantially parallel direction in the longitudinal direction, a second diversion groove **2152** located between two of the first guide grooves **2151**, and a plurality of filling grooves **2153** arranged in a horizontal direction extending between the two first diversion grooves **2151** and the second diversion groove **2152**, wherein two rows of the filling grooves **2153** extend respectively in the two first diversion grooves **2151** and the second diversion groove **2152**.

(103) For example, in this embodiment, the circuit board jointed panel **1100** includes four rows of the circuit boards **11**, and two rows of the circuit boards **11** as a group, and the rigid regions **111** of the two rows of the circuit boards **11** of each group of the circuit boards **11** are located in the middle and are integrally molded, for example, each row of the circuit boards **11** has six circuit boards, and their rigid regions **111** are integrally molded. The molding die **210** has two base jointed panel molding guide grooves **215**, and each of the base jointed panel molding guide grooves **215** has **7** filling grooves **2153** between each of the first diversion groove **2151** and the second diversion groove **2152**, and there are filling grooves **2153** between two adjacent light window molding portions **214**. The molding material **14** flows along the two first diversion grooves **2151** and the middle second diversion groove **2152** from the feeding end **215**A to the terminal end **215**B, and when the dimension and the ratio of the first diversion groove **2151** and the second diversion groove **2152** are within a predetermined range, the molding material **14** can fill each of the filling grooves **2153**, so that the molding material **14** forms the one-piece molding base **1200** after being cured.

(104) In this embodiment of the present application, the one-piece molding base **1200** is integrally molded on two adjacent rows of the circuit boards **11** and two adjacent rows of the photosensitive chips **13** to form a photosensitive assembly jointed panel **1000** and form a light window **122** for providing a light path for each of the photosensitive chips **13** at a position corresponding to the light window molding portion **214**.

(105) As shown in FIGS. **21**A to **24**, it is a schematic view of a manufacturing process of the photosensitive assembly jointed panel **1000** of the camera module **100** according to the preferred embodiment of the present application. As shown in FIG. **21**A, the molding die **210** is in a clamping state, the circuit board **11** to be molded and the solid molding material **14** are ready to be in place, the solid molding material **14** is heated, thereby when the molding material **14** is melted into a fluid state or a semi-solid and semi-fluid state, it is fed to the base jointed panel molding guide groove **215**, and along the first diversion groove **2151** and the second diversion groove **2152** flows forward and fills the filling groove **2153** between two adjacent light window forming portions **214**. In addition, in order to make the molding surface of the first die **211** closely adhere to

the circuit board **11** and the photosensitive chip **13** and facilitate demold, an elastic film layer **219** is further provided between the molding surface of the first die **211** and the circuit board **11** and the photosensitive chip **13**.
(106) As shown in FIGS. **22** and **23**, when the two first diversion grooves **2151**, the second diversion groove **2152**, and the filling groove **2153** of the base jointed panel molding guide groove **215** are all filled with the fluid-like molding material **14**, then the fluid-like molding material **14** is cured and molded into the one-piece molding base **120** integrally molded on two adjacent rows of the circuit board **11** and two rows of the photosensitive chips **13** through a curing process.
(107) As shown in FIG. **24**, after the molding material **14** is cured to form the one-piece molding base **1200**, the demold process of the present application is performed, that is, the die fixing device **230** makes the first die **211** and the second die **212** be far away from each other, so that the light window molding portion **214** is separated from the one-piece molding base **1200**, and two rows of the light windows **122** corresponding to each of the photosensitive chips **13** are formed in the one-piece molding base **1200**.
(108) As shown in FIG. **20**B, the prepared photosensitive assembly jointed panel **1000** can be further cut to obtain a single photosensitive assembly **10**. Each of the photosensitive assemblies **10** includes at least one circuit board **11**, at least one photosensitive chip **13** and the molding base **12** integrally molded on the circuit board **11** and the photosensitive chip **13**. As shown in FIGS. **19**A to **20**B, the rigid regions **111** integrally molded between the two adjacent rows of the circuit boards **11** are separated, so that each of the circuit boards **11** includes a combined rigid region **111** and flexible region **112**. The molding base **12** integrally molds the rigid region **111** of the circuit board **11** and at least a part of the non-photosensitive region **132** of the photosensitive chip **13**, and forms the light window **122** that provides a light path for the photosensitive region **131** of the photosensitive chip **13**.
(109) It is worth mentioning that when each of the photosensitive assemblies **10** produced by cutting the photosensitive assembly jointed panel **1000** is used to make a moving focus camera module, that is, an autofocus camera module, the molding die **210** further provides with a plurality of driver pin groove molding blocks **218**, and each of the driver pin groove molding blocks **218** extends into the filling groove **2153** of the base jointed panel molding guide groove **215**, so as not to affect the flow of the molding material **14** in the three diversion grooves **2151**, **2152** and **2153**, and during the molding process, the fluid-like molding material **14** does not fill the position corresponding to each of the driver pin groove molding blocks **218**, so that after the curing step, a plurality of the light windows **122** and a plurality of driver pin grooves **124** are formed in the one-piece molding base **1200** of the photosensitive assembly jointed panel **1000**, and the molding base **12** of each of the photosensitive assemblies **10** obtained by cutting is configured with the driver pin groove **124**, so that when the moving focus camera module **100** is manufactured, The driver pins can be connected to the circuit board **11** of the photosensitive assembly **10** by means of soldering or conductive adhesive.
(110) It is worth mentioning that the manufacturing method of the photosensitive assembly jointed panel **1000** of the present application is suitable for manufacturing the photosensitive assembly **10** with a small dimension. In the molding process, a width of the bottom end of each of the first diversion grooves **2151** is a, and a width of the bottom end of each of the second diversion grooves **2152** is c. The width a corresponds to the width of the bottom end of the first diversion groove **2151** adjacent to the position where the rigid region **111** and the flexible region **112** of the circuit board **11** are combined, and the width c corresponds to the width of the bottom end of the second diversion groove **2152** on the other side far from the flexible region **112**. When the widths a and c meet the following conditions, that is, 0.2 mm≤a≤1 mm and 0.2 mm≤c≤1.5a, the fluid-like molding material **14** can flow forward along with the two outer first guide grooves **2151** and the middle second guide groove **2152** and fills the entire base jointed panel molding guide groove **215** with the molding material **14** before the molding material **14** is cured. It is worth mentioning that

0.7a≤c≤1.3a is more preferred, for example, in some examples, c=0.8a or c=1a or c=1.2a.

(111) Correspondingly, the molding process of the present application obtains the photosensitive assembly jointed panel **1000**, which includes: one or more rows of the circuit board **11**, one or more rows of the photosensitive chip **13**, and one or more of the one-piece molding base **1200**. Each row of the circuit boards **11** includes one or more of the circuit boards **11** arranged side by side, and each of the circuit boards **11** includes a combined rigid region **111** and flexible region **112**. Each of the one-piece molding bases **1200** is integrally molded in two adjacent rows of the circuit boards **11** and two adjacent rows of the photosensitive chips **13** and forms a light window **122** that provides a light path for each of the photosensitive chips **13**, and the two adjacent rows of the circuit boards **11** are arranged such that their flexible regions **112** are away from each other and their rigid regions **11** are adjacent to each other, so that each of the one-piece molding bases **1200** has two end sides adjacent to the flexible region **112**; wherein a distance between an outer edge **1201** and an inner edge **1202** of a portion **1200**A of the one-piece molding base corresponding to each end side of the one-piece molding base **1200** adjacent to the flexible region **112** is a; the one-piece molding base **1200** extends to a portion **1200**B between the two adjacent rows of the photosensitive chips **13**, and a distance between two inner edges **1204** thereof is c, wherein 0.2 mm≤a≤1 mm, 0.2 mm≤c≤1.5a. Wherein, each of the end sides of the one-piece molding base **1200** corresponds to a combined side of the rigid region **111** and the flexible region **112** of the circuit board **11**, that is, a proximal end side adjacent to the flexible region **112**; the distal end side of the one-piece molding base **1200** corresponding to the circuit board **11** away from the flexible region **112** extends between two adjacent rows of the photosensitive chips **13**.

(112) After the photosensitive assembly jointed panel **1000** is cut, a single photosensitive assembly **10** can be obtained, wherein, in the cutting step, it can be cut on the other side of the one-piece molding base **1200** except for the portion **1200**A of the end side, so that the molding base **12** is obtained, wherein the portions **1200**B corresponding to the molding base between two adjacent rows of the photosensitive chips **13** are also cut.

(113) Accordingly, as shown in FIG. **20**B, the photosensitive assembly **10** obtained after cutting includes the circuit board **11**, the photosensitive chip **13** and the molding base **12**. Wherein, the circuit board **11** includes a combined rigid region **111** and flexible region **112**. The molding base **12** is integrally molded on the circuit board **11** and the photosensitive chip **13** and forms the light window **122** that provides a light path for the photosensitive chip **13**. The circuit board **11** and the photosensitive chip **13** are connected through a series of connecting wires **15**. After cutting the photosensitive assembly jointed panel **1000**, each of the photosensitive assemblies **10** has a first end side without cutting and a second end side obtained by cutting similarly to the above embodiment. A distance between the outer edge **1201** and the inner edge **1202** of a portion **12**A of the molding base corresponding to the first end side of the molding base **12** adjacent to the flexible region **112** is a; the portion **12**B of the molding base **12** corresponding to the opposite second end side of the molding base **12** away from the flexible region **112** has a cutting surface **125**, and a distance between an outer edge **1205** and an inner edge **1204** thereof is b, wherein 0.2 mm≤a≤1 mm, 0.2 mm≤b≤1.5a-0.2 mm.

(114) Based on the consideration of the small-dimension photosensitive assembly **10**, a≤1 mm is selected. And in actual production, it is found that when a<0.2 mm or c<0.2 mm, because the dimensions of the widths a and c are relatively small, the flow velocity and the flow rate of the molding material **14** in the corresponding two first diversion grooves **2151** and the second diversion grooves **2152** are relatively small, so that the base plate molding guide groove **215** cannot be filled with the molding material **14** during the curing time of the molding material. When c>1.5a, that is, when the value of c is greater than 1.5 times of a, in actual production, a part of the base jointed panel molding guide groove **215** cannot be filled, resulting in a defective product. Therefore, in this embodiment of the present application, 0.2 mm≤a≤1 mm and 0.2 mm≤c≤1.5a. The above dimension range of the widths a and c enables the molding material **14** to fill the base

jointed panel molding guide groove **215** in the molding process, thereby avoiding the occurrence of defective products of the photosensitive assembly.

(115) That is to say, in this embodiment of the present application, the molding material **14** can flow forward from the feeding ends **215**A of the three diversion grooves **2151** and **2152** and fill the diversion grooves **2151** and **2152** and the filling groove **2153** of the entire base jointed panel molding guide groove **215**. The molding material **14** can flow from the feeding end **215**A to the terminal end **215**B along with the three diversion grooves **2151** and **2152** before curing. And before the viscosity of the molding material **14** reaches a high value and is cured, the molding material **14** can fill the base jointed panel molding guide groove **215**, thereby preventing the circuit board **11** and the connecting wire **15** between the photosensitive chips **13** is damaged by the molding material **14** with a high viscosity flowing forward. And the fluid in the three diversion grooves **2151** and **2152** flows forward substantially at the same step, so as to prevent the molding material **14** in one diversion groove from flowing to the other diversion groove and obstructing the molding material **14** from flowing forward in the other diversion groove. Moreover, no turbulence or sinuous flow is generated, which causes the connecting wires **15** connecting the circuit board **11** and the photosensitive chip **13** to swing irregularly, resulting in deformation and damage.

(116) As shown in FIG. **21**B, in order to facilitate demold and pressing of the rigid region **111** of the circuit board **11**, the first die **211** further includes a plurality of pressing blocks **216**, and the outer edge **1201** of the molding base **12** and the outer edge of the rigid region **111** of the circuit board **11** will form a pressing edge **1111** with a width of W, that is, in the molding process, the two pressing blocks **216** are pressed on a region of the rigid regions **111** of the two rows of the circuit boards **11**, for example, the pressing distance W may be 0.1˜1 mm. As in a specific example, the pressing distance W may be 0.2 mm. The two pressing blocks **216** are pressed above each group of the flexible regions **112** of the two adjacent rows of the circuit boards **11** to prevent the molding material **14** from flowing to the flexible regions **112**. In addition, the rigid regions **111** of the two adjacent rows of the circuit board **11** are integrally molded to form an integral rigid region jointed panel **110**, and the two pressing blocks **216** are respectively pressed on both end sides of the entire rigid region jointed panel **110**, so as to facilitate the pressing of the first die **211** to the adjacent two rows of the circuit boards **11**.

(117) Accordingly, this embodiment of the present application provides a method for manufacturing the photosensitive assembly **10** of the camera module **100**, which includes the following steps: fixing the circuit board jointed panel **1100** to the second die **212** of the molding die **210**, wherein the circuit board jointed panel **1100** includes one or more rows of circuit boards, and each row of circuit boards includes one or more circuit boards **11** arranged side by side, each of the circuit boards **11** includes a combined rigid region **111** and flexible region **112**, and each of the circuit boards **11** is operatively connected with the photosensitive chip **13**; clamping the second die **212** and a first die **211** through a die fixing device **213** and filling the molten molding material **14** in a base jointed panel molding guide groove **215** in the molding die **210**, wherein the position corresponding to the light window molding portion **214** is prevented from being filled with the molding material **14**; curing the molding material **14** in the base jointed panel molding guide groove **215** to form the one-piece molding base **1200** at a position corresponding to the base jointed panel molding guide groove **215**, wherein the one-piece molding base **1200** is integrally molded on two adjacent rows of the circuit boards **11** and two adjacent rows of the photosensitive chips **13** to form a photosensitive assembly jointed panel **1000** and form a light window **122** for providing a light path for each of the photosensitive chips **13** at a position corresponding to the light window molding portion **214**, wherein the two adjacent rows of the circuit boards **11** are arranged such that their flexible regions **112** are away from each other and their rigid regions **111** are adjacent to each other, wherein the base jointed panel molding guide groove **215** has two first diversion grooves **2151** corresponding to a two end sides of the one-piece molding base **1200** adjacent to the flexible region **112** and a second guide groove **2152** corresponding to a region between the two adjacent

rows of photosensitive chips **13**, and a filling grooves **2153** extending between the first diversion groove **2151** and the second diversion groove **2152** for filling the molding material **14** between two adjacent photosensitive chips **13** in each row of the photosensitive chips **13** and located between two adjacent light window molding portions **214**, wherein a width of the bottom end of the first diversion groove is a, and a width of the bottom end of the second diversion groove is c, wherein the width a corresponds to a distance between an outer edge **1201** and an inner edge **1202** of a portion **1200**A of the one-piece molding base of the each end side of the one-piece molding base **1200** adjacent to the flexible region **112**; wherein the width c corresponds to a distance between two inner edges **1204** of a portion **1200**B of the one-piece molding base extending between the two adjacent rows of the photosensitive chips **13**, wherein 0.2 mm≤a≤1 mm, 0.2 mm≤c≤1.5a. cutting the photosensitive assembly jointed panel **1000** to obtain a plurality of the photosensitive assemblies **10**, wherein each of the photosensitive assemblies **10** includes the circuit board **11**, the photosensitive chip **13** and the molding base **12**, wherein, the molding base **12** is integrally molded on the circuit board **11** and the photosensitive chip **13** and forms the light window **122** that provides a light path for the photosensitive chip **13**.

(118) And, the method may further include the step of cutting a portion of the photosensitive assembly **10** located between the two adjacent rows of the photosensitive chips **13** to obtain a portion **12**B of the molding base corresponding to the opposite other end side of the molding base **12** away from the flexible region **112**, and making a distance between an outer edge **1205** and an inner edge **1204** thereof be b, where 0.2 mm≤b≤1.5a-0.2 mm. That is, the molding base **12** of the photosensitive assembly **10** between the two adjacent rows of the photosensitive chips **13** and the rigid region **111** of the circuit board **11** are adapted to be cut, so that the distal sides of the two adjacent rows of the photosensitive assemblies **10** away from the flexible region **112** are cutting sides, and form cutting surfaces **125** respectively.

(119) The circuit board **11** includes a plurality of electronic components **113** formed in the rigid region **111**, such as mounted by an SMT process, in corresponding two of the first diversion grooves **2151** and the second diversion grooves **2152**, there is no the electronic component **113**, the electronic component **113** can be centrally provided in the filling groove **2153**, so that in the molding process, there will be no obstruction in the two first diversion grooves **2151** and the second diversion groove **2152**, so that the molding material **14** will not be affected to flow forward along with two first diversion grooves **2151** and the second diversion groove **2152**, so that the molding material **14** flow from the feeding end **215**A to the terminal end **215**B in a relatively short time as soon as possible.

(120) In the step of manufacturing a single photosensitive assembly **10**, the photosensitive assembly jointed panel **1000** may be cut to obtain a plurality of independent photosensitive assemblies **10** for manufacturing a single camera module. It is also possible to cut and separate two or more of the photosensitive assemblies **10** integrally connected from the photosensitive assembly jointed panel **1000** to be used to make a separate array camera module, that is, each of the camera modules of the array camera modules each have independent photosensitive assemblies **10**, wherein two or more of the photosensitive assemblies **10** can be respectively connected to a control board of the same electronic device, so that an array camera module made by two or more of the photosensitive assemblies **10** can transmit the images captured by the multiple camera modules to the control board for image information processing.

(121) As shown in FIG. **25**A, the photosensitive assembly **1000** according to another variant embodiment based on the first embodiment of the present application, which includes the photosensitive assembly **1000** obtained by the inventive molding process, including: one or more rows of the circuit board **11**, one or more rows of the photosensitive chips **13**, one or more rows of the protective frame **16**, and one or more of the one-piece molding base **1200**. Each row of the circuit boards **11** includes one or more of the circuit boards **11** arranged side by side, and each of the circuit boards **11** includes a combined rigid region **111** and flexible region **112**. Each of the

protective frames **16** is formed in the photosensitive chip **13** and is located in the non-photosensitive region **132** of the photosensitive chip **13**, that is, it is located outside the photosensitive region **131**, and each of the one-piece molding bases **1200** is integrated molded on a row of the circuit boards **11**, a row of the photosensitive chips **13**, and a row of the protective frames **16** and forms the light windows **122** that provide a light path for each of the photosensitive chips **13**.
(122) That is to say, before the one-piece molding base **1200** is molded, the protective frame **16** is formed on each of the photosensitive chips **13** in advance, which may be formed of another material different from the molding material **14**, for example, it may be glue applied to the non-photosensitive region **132** of the photosensitive chip **13**, or it may be a rigid frame and attached to the non-photosensitive region **132** of the photosensitive chip **13** by glue. Therefore, in the process of molding and forming the one-piece molding base **1200**, the light window molding portion **214** is pressed on the protective frame **16** having a predetermined hardness, when the fluid-like molding material **14** enters the base jointed panel molding guide groove **215**, the fluid-like molding material **14** can be prevented from flowing into the photosensitive region **131** of the photosensitive chip **13**, thereby forming a molding flash. For example, in a specific example, the protective frame **16** is formed of glue, which has a predetermined elasticity and hardness, and can be further implemented to be still sticky after curing, so as to be used for adhering dust particles in the photosensitive assembly **10** of the obtained camera module. More specifically, in some embodiments, the Shore hardness of the protective frame **16** ranges from A50 to A80, and the elastic modulus ranges from 0.1 Gpa to 1 Gpa.
(123) Similarly, a distance between the outer edge **1201** and the inner edge **1202** of a portion **1200**A of the one-piece molding base corresponding to the first end side of the one-piece molding base **1200** adjacent to the flexible region **112** Is a; a distance between an outer edge **1203** and an inner edge **1204** of a portion **1200**B of the one-piece molding base corresponding to the opposite second end of the one-piece molding base **1200** away from the flexible region **112** is c, wherein 0.2 mm≤a≤1 mm, 0.2 mm≤c≤1.5a. Wherein, the first end side of the one-piece molding base **1200** corresponds to a combined side of the rigid region **111** and the flexible region **112** of the circuit board **11**, that is, near the proximal side of the flexible region **112**; the second end side of the one-piece molding base **1200** corresponds to the distal end side of the circuit board **11** away from the flexible region **112**.
(124) After the photosensitive assembly jointed panel **1000** is cut, a single photosensitive assembly **10** can be obtained, as shown in FIG. **25**C, wherein in the cutting step, it can be cut on the one-piece molding base **1200** except for the two wing sides of the first end side and the second end side to obtain the molding base **12**, and the portion **1200**B of the molding base corresponding to the second end side is not cut, so that the photosensitive assembly **10** having a portion **1200**C of the one-piece molding base on a pair of opposite wing sides is obtained.
(125) Accordingly, the photosensitive assembly **10** includes the circuit board **11**, the photosensitive chip **13**, the protective frame **16**, and the molding base **12**. The circuit board **11** includes a combined rigid region **111** and flexible region **112**. The molding base **12** is integrally molded on the circuit board **11**, the photosensitive chip **13** and the protective frame **16** and forms the light window **122** that provides a light path for the photosensitive chip **13**. The circuit board **11** and the photosensitive chip **13** are connected through a series of connecting wires **15**. The protection frame **16** may be located at inner side of the connection wire **15**, or may embed at least a part of the connection wire **15**. A distance between the outer edge **1201** and the inner edge **1202** of a portion **12**A of the molding base corresponding to the first end side of the molding base **12** adjacent to the flexible region **112** is a; a distance between the outer edge **1203** and the inner edge **1204** of a portion **12**B of the molding base corresponding to the opposite second end side of the molding base **12** away from the flexible region **112** is c, where 0.2 mm≤a≤1 mm, 0.2 mm≤c≤1.5a.
(126) As shown in FIG. **25**C, correspondingly, in order to further reduce the dimension of the

photosensitive assembly **10**, at least a part of the photosensitive assembly **10** of the opposite second end side of the molding base **12** away from the flexible region **112** is suitable for being removed to form a cutting surface **125**, so that a distance between the outer edge **1205** and the inner edge **1204** of the remaining part of the molding base **12** after being cut is b, where 0.2 mm≤b≤1.5a-0.2 mm.

(127) As shown in FIG. **25**B, the photosensitive assembly jointed panel **1000** obtained by a molding process according to the variant embodiment of the above-mentioned second example of the present application, which includes: one or more rows of the circuit boards **11**, one or more rows of the photosensitive chips **13**, one or more rows of protective frames **16**, and one or more of the one-piece molding base **1200**. Each of the protective frames **16** is formed on the corresponding photosensitive chip **13**. Each row of the circuit boards **11** includes one or more of the circuit boards **11** arranged side by side, and each of the circuit boards **11** includes a combined rigid region **111** and flexible region **112**. Each of the one-piece molding bases **1200** is integrally molded in two adjacent rows of the circuit boards **11**, two adjacent rows of the photosensitive chips **13**, and two adjacent rows of the protective frames **16** and form a light window **122** that provides a light path for each photosensitive chip **13**, and the two adjacent rows of the circuit boards **11** are arranged such that its flexible regions **112** are far away from each other and its rigid regions **11** are adjacent to each other, so that each of the one-piece molding bases **1200** has two end sides adjacent to the flexible region **112**; wherein, a distance between an outer edge **1201** and an inner edge **1202** of a portion **1200**A of the one-piece molding base corresponding to each end side of the one-piece molding base **1200** adjacent to the flexible region **112** is a; the one-piece molding base **1200** extends between the two adjacent rows of the photosensitive chips **13** and a distance between the two inner edges **1204** thereof is c, wherein 0.2 mm≤a≤1 mm, 0.2 mm≤c≤1.5a. Wherein, each of the end sides of the one-piece molding base **1200** corresponds to a combined side of the rigid region **111** and the flexible region **112** of the circuit board **11**, that is, a proximal end side adjacent to the flexible region **112**; the one-piece molding base **1200** corresponds to the distal side of the circuit board **11** away from the flexible region **112** and extends between two adjacent rows of the photosensitive chips **13**.

(128) After the photosensitive assembly jointed panel **1000** is cut, a single photosensitive assembly **10** can be obtained, and in the cutting step, the other side of the one-piece molding base **1200** except for portion **1200**A of the end side can be cut, so that the molding base **12** is obtained, wherein a portion **1200**B corresponding to the molding base between two adjacent rows of the photosensitive chips **13** are also cut, so that the photosensitive assembly **10** having the portion **1200**C of the one-piece molding base on a pair of opposite wing sides is obtained. A distance between an outer edge **1201** and an inner edge **1202** of a portion **12**A of the molding base corresponding to the first end side of the molding base **12** adjacent to the flexible region **112** is a; the portion **12**B of the molding base corresponding to the opposite second end side of the molding base **12** away from the flexible region **112** has a cutting surface **125** and a distance between an outer edge **1205** and an inner edge **1204** thereof is b, wherein 0.2 mm≤a≤1 mm, 0.2 mm≤b≤1.5a-0.2 mm, as shown in FIG. **25**C.

(129) As shown in FIG. **26**A to FIG. **27**, the molding process of the jointed paneling operation can also be used to make a photosensitive assembly **10** having two or more of the light windows **122**, wherein such a photosensitive assembly **10** can be used to fabricate an array camera module that shares a substrate. That is to say, taking the photosensitive assembly **10** for making a dual-camera module as an example, during the molding process of each circuit board **11** of the circuit board jointed panel **1100**, one circuit board substrate **111** is correspondingly provided with two light window molding portions **214**, so that after the molding process and cutting are completed, each of the circuit boards **11** forms a molding base **12** having two of the light windows **122** sharing one circuit board **11**, and correspondingly mounts the two photosensitive chips **13** and two lenses **30**. In addition, the circuit board **11** can be connected to a control board of an electronic device. In this way, the array camera module manufactured in this embodiment can transmit images captured by multiple camera modules to the control board for image information processing.

(130) As shown in FIGS. **28**A to **32** is a method for manufacturing a photosensitive assembly according to a third preferred embodiment of the present application. In this preferred embodiment of the present application, each of the base jointed panel molding guide grooves **215** has a first diversion groove **2151** and a second diversion groove **2152** substantially parallel to each other, and a plurality of filling grooves **2153** extends between the first diversion groove **2151** and the second diversion groove **2152**, wherein the filling groove **2153** is formed between two adjacent light window molding portions **214**, as shown in the figure, the base jointed panel molding guide groove **215** has seven filling grooves **2153**, and six light window molding portions **214** are located between two adjacent filling grooves **2153**. The molding material **14** flows along with the first diversion groove **2151** and the second diversion groove **2152** from the feeding end **215**A toward the terminal end **215**B. The second diversion groove **2152** is larger in dimension, and when the first die **211** and the second die **212** are clamped, one or more flow retarding parts **114** are provided on the circuit board **11** in the second diversion groove **2152**, and the flow retarding parts **114** slow down a flow rate of the molding material **14** in the second diversion groove **2152**, so that the fluid-like molding material **14** can be filled in each of the filling grooves **2153** along the two diversion grooves **2151** and **2152** within a curing time T, thereby the one-piece molding base **1200** is formed after the molding material **14** is cured.
(131) As shown in FIG. **32**, it is a schematic view of the manufacturing process of the photosensitive assembly jointed panel **1000** of the camera module **100** according to this preferred embodiment of the present application. The molding die **210** is in a clamping state, the circuit board **11** to be molded and the solid molding material **14** are ready to be put into place, the solid molding material **14** is heated, so that when the molding material **14** is melted into a fluid state or a semi-solid and semi-fluid state, it is fed to the base jointed panel molding guide groove **215**, flows forward along with the first diversion groove **2151** and the second diversion groove **2152**, and fills into the filling grooves **2153** between two adjacent light window molding portions **214**, wherein the retarding part **114** located in the second diversion groove **2152** blocks the flow of the fluid in the second diversion groove **2152**, so that The flow velocity of the molding material **14** in the second diversion groove **2152** is reduced. In this way, when the base molding plate molding guide groove **215** can be completely filled with the fluid-like molding material **14**, the fluid-like molding material **14** is cured and molded into the one-piece molding base **1200** integrally formed in each row of the circuit boards **11** and each row of the photosensitive chips **13** through a curing process.
(132) It is worth mentioning that in the molding process, a width of the bottom end of the first diversion groove **2151** is a, and a width of the bottom end of the second diversion groove **2152** is d. The width a corresponds to a width of the bottom end of the first diversion groove **2151** adjacent to the position where the rigid region **111** and the flexible region **112** of the circuit board **11** are combined, and the width d corresponds to a width of the bottom end of the second diversion groove **2152** on the other side far from the flexible region **112**. When the widths a and d meet the following conditions, 0.2 mm≤a≤1 mm and d>1.5a, accordingly, in the molding process, since the flow retarding part **114** is provided in the second diversion groove **2152**, so the purpose of filling the base jointed panel molding guide groove **215** in the molding process can also be achieved.
(133) Correspondingly, the molding process of the present application yields the photosensitive assembly jointed panel **1000**, which includes: one or more rows of the circuit board **11**, one or more rows of the photosensitive chip **13**, and one or more of the one-piece molding base **1200**. Each row of the circuit boards **11** includes one or more of the circuit boards **11** arranged side by side, and each of the circuit boards **11** includes a combined rigid region **111** and flexible region **112**. Each of the one-piece molding bases **1200** is integrally molded on a row of the circuit boards **11** and a row of the photosensitive chips **13** and forms the light windows **122** that provide a light path for each of the photosensitive chips **13**. Wherein, a distance between the outer edge **1201** and the inner edge **1202** of the portion **1200**A of the one-piece molding base corresponding to the first end side of the one-piece molding base **1200** adjacent to the flexible region **112** is a; a distance between the outer

edge **1203** and the inner edge **1204** of the portion **1200**B of the one-piece molding base corresponding to the opposite second end side of the one-piece molding base **1200** away from the flexible region **112** is d, where 0.2 mm≤a≤1 mm, and d>1.5a. The second end side of the one-piece molding base **1200** corresponds to the circuit board **11** on the distal side of the circuit board **11** away from the flexible region **112** and further includes one or more flow retarding part **114** provided on the rigid region **111** thereof, and the one-piece molding base **1200** embeds the flow retarding part **114** integrally.

(134) After the photosensitive assembly jointed panel **1000** is cut, a single photosensitive assembly **10** can be obtained, wherein the photosensitive assembly **10** includes the circuit board **11**, the photosensitive chip **13**, and the molding base **12**. Wherein, the circuit board **11** includes a combined rigid region **111** and flexible region **112**. The molding base **12** is integrally molded on the circuit board **11** and the photosensitive chip **13** and forms the light window **122** that provides a light path for the photosensitive chip **13**. The circuit board **11** and the photosensitive chip **13** are connected through a series of connecting wires **15**. The distance between the outer edge **1201** and the inner edge **1202** of the portion **1200**A of the one-piece molding base corresponding to the first end side of the one-piece molding base **1200** adjacent to the flexible region **112** is a; the distance between the outer edge **1203** and the inner edge **1204** of the portion **1200**B of the one-piece molding base corresponding to the opposite second end side of the one-piece molding base **1200** away from the flexible region **112** is d, where 0.2 mm≤a≤1 mm, and d>1.5a, and the portion **12**B of the molding base of the opposite second end side of the molding base **12** away from the flexible region **112** is embedded with the flow retarding part **114**.

(135) As shown in FIG. **31**, correspondingly, in order to further reduce the dimension of the photosensitive assembly **10**, at least a part of the photosensitive member **10** of the opposite second end side of the molding base **12** away from the flexible region **112** is suitable for being removed, such as cutting with a knife or being abraded, to form a cutting surface **125**, and the distance between the outer edge **1205** and the inner edge **1204** of the remaining part of the molding base **12** after being cut is b, wherein 0.2 mm≤b≤d-0.2 mm. Preferably, a portion of the molding base **12** in which the flow retarding part **114** is embedded is removed, and a part of the corresponding circuit board **11** is also removed, preferably, 0.2 mm≤b≤1.5a-0.2 mm.

(136) Correspondingly, this embodiment of the present application provides the following method for manufacturing a photosensitive assembly of a camera module, which includes the following steps: fixing a circuit board jointed panel **1100** to a second die **212** of a molding die **210**, wherein the circuit board jointed panel **1100** includes one or more rows of circuit boards, and each row of circuit boards includes one or more circuit boards **11** arranged side by side, each of the circuit boards **11** includes a combined rigid region **111** and flexible region **112**, and each of the circuit boards **11** is operatively connected with a photosensitive chip **13**; clamping the second die **212** and a first die **211** and filling the molten molding material **14** in a base jointed panel molding guide groove **215** in the molding die **210**, wherein the position corresponding to a light window molding portion **214** is prevented from being filled with the molding material **14**; curing the molding material **14** in the base jointed panel molding guide groove **215** to form a one-piece molding base **1200** at a position corresponding to the base jointed panel molding guide groove **215**, wherein the one-piece molding base **1200** is integrally molded on corresponding each row of the circuit boards **11** and each row of the photosensitive chips **13** to form a photosensitive assembly jointed panel **1100** and form a light window **122** for providing a light path for each of the photosensitive chips **13** at a position corresponding to the light window molding portion **214**, wherein the base jointed panel molding guide groove **215** has a first diversion groove **2151** corresponding to a first end side of the one-piece molding base **1200** adjacent to the flexible region **112** and a second diversion groove **2152** corresponding to the one-piece molding base **1200** away from the flexible region **112**, and a filling grooves **2153** extending between the first diversion groove **2151** and the second diversion groove **2152** and for filling the molding material **14** between two adjacent photosensitive

chips **13** in each row of the photosensitive chips **13** and located between two adjacent light window molding portions **214**, wherein a width of the bottom end of the first diversion groove is a, and a width of the bottom end of the second diversion groove is d, wherein 0.2 mm≤a≤1 mm, d>1.5a, and a flow retarding part is provide on the rigid region **111** corresponding to a second end side of the one-piece molding base **1200** away from the flexible region **112**, and the flow retarding part **114** is embedded by the one-piece molding base **1200**; cutting the photosensitive assembly jointed panel **1000** to obtain a plurality of the photosensitive assemblies **10**, wherein each of the photosensitive assemblies **10** includes the circuit board **11**, the photosensitive chip **13** and the molding base **12**, wherein, the molding base **12** is integrally molded on the circuit board **11** and the photosensitive chip **13** and forms the light window **122** that provides a light path for the photosensitive chip **13**. And preferably, in the cutting step, a portion of the molding base **12** in which the flow retarding part **114** is embedded is removed.

(137) The photosensitive assembly **10** of the camera module **100** according to the fourth embodiment of the present application and a manufacturing process thereof are shown in FIGS. **33**A to **37**. In this embodiment, a photosensitive assembly jointed panel plate **1000** is also made by a jointed panel operation, and then the photosensitive assembly **10** is cut. In this embodiment, the difference from the third embodiment is that the two adjacent rows of circuit boards can arrange the rigid regions **111** adjacently, and keep the corresponding flexible regions **112** far away. More preferably, the rigid regions **111** of the two adjacent rows of circuit boards are integrally molded so that the middle of the two adjacent rows of circuit boards forms an integral rigid region.

(138) Accordingly, more specifically, the molding die **210** forms a molding cavity **213** when clamping, and provides a plurality of light window molding portions **214** and one or more base jointed panel molding guide grooves **215**, each of the bases jointed panel molding guide grooves **215** includes a first diversion groove **2151** arranged at two ends in a substantially parallel direction in the longitudinal direction, and a second diversion groove **2152** located between two first guide grooves **2151** and a plurality of filling grooves **2153** arranged laterally extending between the first diversion groove **2151** and the second diversion groove **2152**, wherein two rows of the filling grooves **2153** extend between the two first diversion grooves **2151** and the second diversion groove **2152**.

(139) In this embodiment, in the molding process, the second diversion groove **2152** having a larger dimension is provided with a flow retarding part **114** to reduce the flow velocity of the fluid in the second diversion groove **2152**, so that the molding material **14** flows from the feeding end **215**A to the terminal end **215**B along with two of the first diversion guides **2151** and the second diversion guide **2152** in the middle, and the molding material **14** can fill each of the filling grooves **2153**, so that the one-piece molding base **1200** is formed after the molding material **14** is cured.

(140) In this embodiment of the present application, the one-piece molding base **1200** is integrally molded on two adjacent rows of the circuit boards **11** and two adjacent rows of the photosensitive chips **13** to form a photosensitive assembly jointed panel **1000** and form a light window **122** for providing a light path for each of the photosensitive chips **13** at a position corresponding to the light window molding portion **214**. The flow retarding part **114** is provided on the rigid region **111** of the circuit board **11** between the two rows of the photosensitive chips **13**, so that the flow of the molding material **14** in the second diversion groove **2152** is properly blocked.

(141) As shown in FIG. **37**, the molding die **210** is in a clamping state, the circuit board **11** to be molded and the solid molding material **14** are ready to be in place, and the solid molding material **14** is heated, so that when the molding material **14** is melted into a fluid state or a semi-solid and semi-fluid state, it is fed to the base jointed panel molding guide groove **215**, and along the first diversion groove **2151** and the second diversion groove **2152** flows forward and fills the filling groove **2153** between two adjacent light window molding portions **214**, wherein the flow retarding part **114** is provided to reduce the flow velocity of the molding material **14** in the second diversion groove **2152**.

(142) As shown in FIGS. **34** to **36**, the prepared photosensitive element jointed panel **1000** can be further cut, so as to obtain a single photosensitive assembly **10**. Each of the photosensitive assemblies **10** includes at least one circuit board **11**, at least one photosensitive chip **13** and the molding base **12** integrally molded on the circuit board **11** and the photosensitive chip **13**. The rigid regions **111** integrally molded between the two adjacent rows of the circuit boards **11** are separated, and preferably, a portion of the molding base **12** in which the retarder **114** is embedded is removed.
(143) In the molding process, the width of the bottom end of each of the first diversion grooves **2151** is a, and the width of the bottom end of each of the second diversion grooves **2152** is c. The width a corresponds to the width of the bottom end of the first diversion grooves **2151** adjacent to the position where the rigid region **111** and the flexible region **112** of the circuit board **11** are combined, and the width d corresponds to the width of the bottom end of the second diversion groove **2152** away from the other side of the flexible region **112**. The widths a and d meet the following conditions, that is, 0.2 mm≤a≤1 mm and d>1.5a. The flow retarding part **114** is provided on the rigid region **111** of the circuit board **11** in the second diversion groove **2152**, the fluid-like molding material **14** can flow forward along the two outer first diversion grooves **2151** and the second diversion groove **2152** in the middle, and before the molding material **14** is cured, the entire base jointed panel molding guide groove **215** is filled with the molding material **14**.
(144) Correspondingly, the photosensitive assembly jointed panel **1000** obtained according to the molding process of the present application includes: one or more rows of the circuit board **11**, one or more rows of the photosensitive chip **13**, and one or more one-piece molding base **1200**. Each row of the circuit boards **11** includes one or more of the circuit boards **11** arranged side by side, and each of the circuit boards **11** includes a combined rigid region **111** and flexible region **112**. Each of the one-piece molding bases **1200** is integrally molded on two adjacent rows of the circuit boards **11** and two adjacent rows of the photosensitive chips **13** and forms a light window **122** that provides a light path for each of the photosensitive chips **13**, and the two adjacent rows of the circuit boards **11** are arranged such that their flexible regions **112** are away from each other and their rigid regions **111** are adjacent to each other, so that each of the one-piece molding bases **1200** has two end sides adjacent to the flexible region **112**; wherein the distance between an outer edge **1201** and an inner edge **1202** of a portion **1200**A of the one-piece molding base corresponding to each end side of the one-piece molding base **1200** adjacent to the flexible region **112** is a; the one-piece molding base **1200** extends between the two adjacent rows of the photosensitive chips **13** and the distance between the two inner edges **1204** thereof is d, where 0.2 mm≤a≤1 mm, d>1.5a, and the flow retarding part **114** is provided on the rigid region **111** of the circuit board **11** between the two adjacent rows of the photosensitive chips **13**, and the flow retarding part **114** is embedded integrally by the one-piece molding base **1200**.
(145) Accordingly, as shown in FIG. **36**, the photosensitive assembly **10** obtained after cutting includes the circuit board **11**, the photosensitive chip **13** and the molding base **12**. After cutting the photosensitive assembly jointed panel **1000**, each of the photosensitive assemblies **10** has a first end side without cutting and a second end side obtained by cutting similarly to the above embodiment. The distance between the outer edge **1201** and the inner edge **1202** of the portion **12**A of the molding base corresponding to the first end side of the molding base **12** adjacent to the flexible region **112** is a; the portion **12**B of the molding base corresponding to the opposite second end side of the molding base **12** away from the flexible region **112** forms a cutting surface **125** and the distance between the outer edge **1205** and the inner edge **1204** thereof is b, wherein, 0.2 mm≤a≤1 mm, 0.2 mm≤b≤d-0.2 mm, preferably, 0.2 mm≤b≤1.5a-0.2 mm.
(146) That is to say, in this embodiment of the present application, since the flow retarding part **114** is provided in the second diversion groove **2152**, the fluid flow velocity in the second diversion groove **2152** is reduced, the molding material **14** can flow forward from the feeding ends **215**A of the three guide grooves **2151** and **2152** and fill the guide grooves **2151** and **2152** of the entire base jointed panel molding guide groove **215** and the filling groove **2153**. The molding material **14** can

flow from the feed end **215**A to the terminal end **215**B along the three diversion grooves **2151** and **2152** before curing. Before the viscosity of the molding material **14** reaches a high value and is cured, the molding material **14** can fill the base jointed panel molding guide groove **215**, thereby preventing the connecting wire **15** between the circuit board **11** and the photosensitive chip **13** is damaged by the molding material **14** having a high viscosity and flowing forward. And the fluid in the three diversion grooves **2151** and **2152** flows forward substantially at the same step, so as to prevent the molding material **14** in one certain diversion groove from flowing to another diversion groove and obstructing the molding material **14** from flowing forward in another diversion groove. Moreover, no turbulence or sinuous flow is generated, which causes the connecting wires **15** connecting the circuit board **11** and the photosensitive chip **13** to swing irregularly, resulting in deformation and damage.

(147) Those skilled in the art should understand that the embodiment of the present application shown in the above description and the accompanying drawings are merely examples and do not limit the present application. The object of the present application has been completely and effectively achieved. The function and structural principle of the present application have been shown and explained in the examples, and the embodiments of the present application may have any deformation or modification without departing from the principle.

Claims

1. A photosensitive assembly of a camera module, the photosensitive assembly comprising: a circuit board including a combined rigid region and flexible region; a photosensitive chip; and a molding base, wherein the molding base is integrally molded on the circuit board and the photosensitive chip and forms a light window for providing a light path for the chip, and wherein a distance between an outer edge and an inner edge of a portion of the molding base corresponding to a first end side of the molding base adjacent to the flexible region is a, distance between an outer edge and an inner edge of a portion of the molding base corresponding to an opposite second end side of the molding base away from the flexible region is c, and 0.2 mm≤a≤1 mm, 0.2 mm≤c≤1.5a; wherein the portion of the molding base corresponding to the opposite second end side away from the flexible region is adapted to be cut, so that a distance between an outer edge and an inner edge of a remaining part of the molding base after being cut is b, and 0.2 mm≤b≤1.5a-0.2 mm.
2. The photosensitive assembly according to claim 1, wherein 0.7a≤c<1.3a.
3. The photosensitive assembly according to claim 1, wherein the circuit board further includes a plurality of electronic components, wherein the plurality of electronic components are provided at at least one wing side that are at sides of the photosensitive assembly on the rigid region except for end sides that are adjacent to the flexible region and away from the flexible region, and wherein the molding base integrally embeds the plurality of electronic components.
4. The photosensitive assembly according to claim 1, wherein at the first end side of the molding base adjacent to the flexible region, a pressing distance W for a pressing block of a molding die in a molding process to facilitate pressing on the rigid region is between the outer edge of the molding base and an outer edge of the rigid region, and the pressing distance W has a value ranging from 0.1˜1 mm.
5. The photosensitive assembly according to claim 1, further comprising a filter assembly and a filter assembly lens holder, wherein the filter assembly is assembled to the filter assembly lens holder, and the filter assembly lens holder is assembled on a top side of the molding base.
6. The photosensitive assembly according to claim 1, further comprising a protective frame provided on the photosensitive chip, wherein the molding base is integrally molded on the circuit board, the photosensitive chip and the protective frame.
7. A photosensitive assembly jointed panel of a camera module, the photosensitive assembly jointed panel comprising: one or more rows of circuit boards, each row of circuit boards including

one or more circuit boards arranged side by side, each of the circuit boards including a combined rigid region and flexible region; one or more rows of photosensitive chips; and one or more one-piece molding bases, each of which is integrally molded on a row of the circuit boards and a row of the photosensitive chips and forms a light window for providing a light path for each of the photosensitive chips, wherein a distance between an outer edge and an inner edge of a portion of the one-piece molding base corresponding to a first end side of the one-piece molding base adjacent to the flexible regions is a, a distance between an outer edge and an inner edge of a portion of the one-piece molding base corresponding to an opposite second end side of the one-piece molding base away from the flexible regions is c, and 0.2 mm ≤a≤1 mm, 0.2 mm≤c≤1.5a; wherein the portion of the molding base corresponding to the opposite second end side away from the flexible region is adapted to be cut, so that a distance between an outer edge and an inner edge of a remaining part of the molding base after being cut is b, and 0.2 mm≤b≤1.5a-0.2 mm.

8. The photosensitive assembly jointed panel according to claim 7, wherein the photosensitive assembly jointed panel includes a plurality of rows of the circuit boards, a plurality of rows of the photosensitive chips, and a plurality of the one-piece molding bases, and wherein in two adjacent rows of the circuit boards, the opposite second end side of the one-piece molding base on one of the two adjacent rows of the circuit boards away from the flexible regions faces towards the first end side of the one-piece molding base adjacent to the flexible regions of another of the two adjacent rows of the circuit boards.

9. The photosensitive assembly jointed panel according to claim 7, wherein 0.7a≤c≤1.3a.

10. The photosensitive assembly jointed panel according to claim 7, wherein the rigid region of each of the circuit boards has a pressing distance W for a pressing block of a first molding die in a molding process to facilitate pressing on the first end side adjacent to the flexible region, and the pressing distance W has a value ranging from 0.1˜1 mm.

11. The photosensitive assembly jointed panel according to claim 7, wherein the rigid regions of a row of the circuit boards are integrally molded to form an integral rigid region jointed panel.

12. The photosensitive assembly jointed panel according to claim 7, wherein each of the one or more one-piece molding bases is integrally molded on two rows of adjacent circuit boards and two rows of adjacent photosensitive chips, and the two adjacent rows of circuit boards are arranged such that the flexible regions thereof are spaced apart from each other and the rigid regions thereof are adjacent to each other, thereby each of the one or more one-piece molding bases has two end sides adjacent to the flexible region regions.

13. The photosensitive assembly jointed panel according to claim 12, wherein the rigid region of each of the circuit boards has a pressing distance W for a pressing block of a first molding die in a molding process to facilitate pressing on the first end side adjacent to the flexible region, and the pressing distance W has a value ranging from 0.1˜1 mm.

14. The photosensitive assembly jointed panel according to claim 12, wherein the rigid regions of the two adjacent rows of circuit boards are integrally molded to form an integral rigid region jointed panel.

15. A camera module, comprising: a lens; a circuit board including a combined rigid region and flexible region; a photosensitive chip; and a molding base, wherein the molding base is integrally molded on the circuit board and the photosensitive chip and forms a light window for providing a light path for the photosensitive chip, wherein the lens is located on a photosensitive path of the photosensitive chip, and wherein a distance between an outer edge and an inner edge of a portion of the molding base corresponding to a first end side of the molding base adjacent to the flexible region is a, a distance between an outer edge and an inner edge of a portion of the molding base corresponding to an opposite second end side of the molding base away from the flexible region is c, and 0.2 mm≤a≤1 mm, 0.2 mm≤c≤1.5a; wherein the portion of the molding base corresponding to the opposite second end side away from the flexible region is adapted to be cut, so that a distance between an outer edge and an inner edge of a remaining part of the molding base after being cut is

b, and 0.2 mm≤b≤1.5a-0.2 mm.
16. The camera module according to claim 15, wherein 0.7a≤c≤1.3a.
17. The camera module according to claim 15, wherein the circuit board further includes a plurality of electronic components, wherein the plurality of electronic components are provided at at least one wing side that are at sides of the photosensitive assembly on the rigid region except for end sides that are adjacent to the flexible region and away from the flexible region, and wherein the molding base integrally embeds the plurality of electronic components.
18. The camera module according to claim 15, wherein the molding base integrally molded on the circuit board and the photosensitive chip is cut from a one-piece molding base made in a jointed panel molding process, and the opposite second end side of the molding base away from the flexible region corresponds to a cutting side.
19. The camera module according to claim 15, further comprising a filter assembly and a filter assembly lens holder, wherein the filter assembly is assembled to the filter assembly lens holder, and the filter assembly lens holder is assembled on a top side of the molding base.